

M2 de Mathématiques Fondamentales

Algebraic Number Theory I

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Disclaimer

All pictures of mathematicians in these course notes are taken from Wikipedia.

Chapter 1

Number fields and their rings of integers

1.1 What are rings of integers?

1.1.1 Integrality

Definition 1.1.1

1. Let $A \subset B$ be an inclusion of domains. An element $x \in B$ is said to be **integral over** A if it is a root of a monic polynomial with coefficients in A . The ring B is said to be **integral over** A if any element of B is integral over A .
2. A complex number that is integral over $\mathbb{Z}$ is called an **algebraic integer**. The set of algebraic integers is denoted $\overline{\mathbb{Z}}$.

Example 1.1.2. 1. For $d \in \mathbb{Z}$, the ring $\mathbb{Z}[\sqrt{d}]$ is contained in $\overline{\mathbb{Z}}$. Indeed, any element x of $\mathbb{Z}[\sqrt{d}]$ can be written as $a + b\sqrt{d}$ with $a, b \in \mathbb{Z}$, and hence satisfies the equation $(x - a)^2 - b^2 d = 0$.

2. For $d \in \mathbb{Z}$ a square free integer congruent to 1 modulo 4, the ring $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ is contained in $\overline{\mathbb{Z}}$. Indeed, any element x of $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ can be written as $a + b\frac{1+\sqrt{d}}{2}$ with $a, b \in \mathbb{Z}$, and hence satisfies the equation

$$x^2 - (2a + b)x + a^2 + ab + b^2 \frac{1-d}{4} = 0.$$

3. For $n \geq 1$, the complex number $e^{2i\pi/n}$ is an algebraic integer because it satisfies the equation $(e^{2i\pi/n})^n = 1$.

Proposition 1.1.3

Let A be a unique factorization domain with fraction field K . Then:

$$\{x \in K : x \text{ integral over } A\} = A.$$

In particular, a rational number is an algebraic integer if, and only if, it is an integer. Equivalently, $\overline{\mathbb{Z}} \cap \mathbb{Q} = \mathbb{Z}$.

Proof. Of course, any $x \in A$ is integral over A since it is a root of the polynomial $X - x$. Now take an element $x = p/q \in K$, with $p \in A$, $q \in A \setminus \{0\}$ and $p \wedge q = 1$, and assume it is a root of some monic polynomial $f = X^n + \sum_{i=0}^{n-1} a_i X^i \in A[X]$. Then:

$$p^n + q \sum_{i=0}^{n-1} a_i p^i q^{n-i-1} = 0$$

Hence q divides p^n . But p and q are coprime. Hence $q = 1$ and $x \in A$. $\square$

Let us now provide a useful criterion to prove that a complex number is an algebraic integer, and more generally to prove that an element of some ring satisfies some monic polynomial equation.

Proposition 1.1.4

Let $A \subset B$ be an inclusion of domains. Fix a finite number of elements $b_1, \dots, b_n \in B$, assumed to be not all zero, and set $\Gamma := b_1 A + \dots + b_n A \subset B$. Consider an element $b \in B$ such that $b\Gamma \subset \Gamma$, and for $1 \leq i \leq n$, write:

$$bb_i = \sum_{j=1}^n a_{ij} b_j$$

with $a_{ij} \in A$. Introduce the monic polynomial:

$$\chi(X) := \det(XI_n - (a_{ij})_{1 \leq i, j \leq n}) \in A[X].$$

Then $\chi(b) = 0$.

Remark 1.1.5. In the proof, we will use without further explanation the following basic result of linear algebra: for any matrix $M \in \mathcal{M}_n(A)$ with coefficients in a ring A , we have ${}^t(\text{Com}(M))M = \det M \cdot I_n$ where $\text{Com}(M)$ stands for the comatrix of M .

Proof. Consider the matrix $M := bI_n - (a_{ij})_{1 \leq i, j \leq n} \in \mathcal{M}_n(B)$. We then have:

$$M \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0,$$

and hence:

$$\det M \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = {}^t(\text{Com}(M))M \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0.$$

We deduce that $\det M = 0$, and hence that $\chi(b) = 0$. $\square$

Example 1.1.6. Take $A = \mathbb{Z}$, $B = \mathbb{C}$ and $b_1 = 1$, $b_2 = \sqrt[3]{2}$, $b_3 = \sqrt[3]{4}$ and $b = 1 + 3\sqrt[3]{2} - \sqrt[3]{4}$. We have $b\Gamma \subset \Gamma$, and we can compute:

$$\chi = \begin{vmatrix} X-1 & 2 & -6 \\ -3 & X-1 & 2 \\ 1 & -3 & X-1 \end{vmatrix} = X^3 - 3X^2 + 21X + 31.$$

We deduce that b is a root of χ , and hence that b is an algebraic integer.

Corollary 1.1.7

Let $A \subset B$ be an inclusion of domains and take $x_1, \dots, x_r \in B$. Then $x_1, \dots, x_r$ are all integral over A if, and only if, the ring $A[x_1, \dots, x_r]$ is finitely generated as an A -module.

Proof. Assume first that the ring $A[x_1, \dots, x_r]$ is finitely generated as an A -module, and let $b_1, \dots, b_n$ be some generators. For each i , we may apply the previous proposition to $b = x_i$ (with $\Gamma = b_1A + \dots + b_nA = B$) and get that x_i is a root of some monic polynomial in $A[X]$. In other words, all the x_i are integral over A .

Conversely, assume that $x_1, \dots, x_r$ are all integral over A . We want to prove that $A[x_1, \dots, x_r]$ is finitely generated as an A -module. By induction on r , it suffices to prove the case $r = 1$. In that case, we let $\mu \in A[X]$ be a monic polynomial such that $\mu(x_1) = 0$ and we denote d its degree. Given $y = P(x_1) \in A[x_1]$ with $P \in A[X]$, we write $P = \mu Q + R$ with $Q, R \in A[X]$ and $\deg R < d$. Then $y = R(x_1) \in \sum_{i=0}^{d-1} Ax_1^i$. Hence $A[x_1] = \sum_{i=0}^{d-1} Ax_1^i$ is finitely generated as an A -module. $\square$

This criterion has important and useful consequences:

Corollary 1.1.8

Let $A \subset B$ be an inclusion of domains. The subset $\overline{A}$ of B consisting of those element that are integral over A is a subring of B . In particular, the set $\overline{\mathbb{Z}}$ is a subring of $\mathbb{C}$.

Proof. Consider two elements $x, y \in \overline{A}$ and let $C = \mathbb{Z}[x, y]$ be the subring of B spanned by x and y . By Corollary 1.1.7, C is a finitely generated A -module. But since $C = \mathbb{Z}[x, y, xy, x + y]$, we also deduce that xy and $x + y$ lie in $\overline{A}$. Hence $\overline{A}$ is a subring of B . $\square$

Corollary 1.1.9

Let $A \subset B \subset C$ be a tower of inclusions of domains. Assume that C is integral over B and that B is integral over A . Then C is integral over A .

Proof. Let $c \in C$. Let $\mu \in B[X]$ be a monic polynomial with $\mu(c) = 0$. Let B' be the sub- A -algebra of B generated by the coefficients of μ . By Corollary 1.1.7, the A -module B' and the B' -module $B'[c]$ are finitely generated. We deduce that $B'[c]$ is finitely generated as an A -module, and hence, applying Corollary 1.1.7 again, c is integral over A . $\square$

We finish this section by introducing the fundamental notions of integral closure and ring of integers:

Definition 1.1.10

1. Let $A \subset B$ be an inclusion of domains. The ring:

$$\overline{A} := \{x \in B : x \text{ integral over } A\}$$

is called the **integral closure of A in B** .

2. Let A be a domain with fraction field K . The **integral closure of A** is the integral closure of A in K . When it is equal to A itself, we say that A is **integrally closed**.
3. Let K be a **number field**, that is a finite extension of $\mathbb{Q}$. The integral closure of $\mathbb{Z}$ in K is called the **ring of integers of K** and in denote $\mathcal{O}_K$. We have $\mathcal{O}_K = \overline{\mathbb{Z}} \cap K$.

Example 1.1.11. 1. By Proposition 1.1.3, unique factorization domains are integrally closed.

2. For $n \in \mathbb{Z}$ a square free integer congruent to 1 modulo 4, the ring $\mathbb{Z}[\sqrt{d}]$ is not integrally closed because $\frac{1+\sqrt{d}}{2}$ is an algebraic integer.

Remark 1.1.12. Let $A \subset B \subset C$ be a tower of domains. Let B' be the integral closure of A in B and let C' be the integral closure of B' in C . Then, by Corollary 1.1.9, the ring C' coincides with the integral closure of A in C . In particular, rings of integers of number fields are integrally closed.

In algebraic number theory, one very often works in the following framework:

Framework 1.1.13

- A an integrally closed domain with fraction field K ;
- B the integral closure of A in a finite separable extension L of K ;

$$\begin{array}{ccc} L & \supset & B \\ | & & | \\ K & \supset & A. \end{array}$$

A particularly interesting situation from an arithmetic perspective arises when $K = \mathbb{Q}$, $A = \mathbb{Z}$ and $B = \mathcal{O}_L$.

In general, it can be quite hard to compute integral closures and rings of integers. In the next section, we provide some methods to do so.

1.1.2 How to compute integral closures and rings of integers?

In this whole section, we place ourselves in the general Framework 1.1.13, and from time to time, when indicated, we restrict ourselves to the arithmetically relevant case $K = \mathbb{Q}$, $A = \mathbb{Z}$ and $B = \mathcal{O}_L$.

Let us start by computing the rings of integers of quadratic number fields. To do so, we first prove the following useful proposition:

Proposition 1.1.14

Let $z \in L$. Then $z \in B$ if, and only if, its minimal polynomial has coefficients in A .

Proof. The converse direction is obvious. Let us therefore assume that z is integral over A and let $P \in A[X]$ be a monic polynomial such that $P(z) = 0$. Let $z_1 = z, \dots, z_r$ be the conjugates of z , that is the roots of the minimal polynomial μ of z in a fixed algebraic closure $\bar{L}$ of L . They are all roots of P , and hence they are all integral over A . Moreover, the coefficients of μ all belong

to $A[z_1, \dots, z_r]$. Hence, by Corollary 1.1.8, they are all integral over A . Since A is integrally closed, we deduce that they belong to A . $\square$

Corollary 1.1.15

Take $K = \mathbb{Q}$, $A = \mathbb{Z}$ and $L := \mathbb{Q}(\sqrt{d})$ for some square-free integer $d \in \mathbb{Z} \setminus \{0, 1\}$. Then:

$$\mathcal{O}_L = \begin{cases} \mathbb{Z}[\frac{1+\sqrt{d}}{2}] & \text{if } d \equiv 1 \pmod{4}, \\ \mathbb{Z}[\sqrt{d}] & \text{if } d \equiv 2, 3 \pmod{4}. \end{cases}$$

Proof. Take an element $x = a + b\sqrt{d} \in L$ with $a \in \mathbb{Q}$ and $b \in \mathbb{Q}^\times$. Its minimal polynomial is $X^2 - 2aX + a^2 - db^2$, and hence x is an algebraic integer if, and only if, $2a$ and $a^2 - db^2$ both belong to $\mathbb{Z}$. The latter condition implies that $d(2b)^2 = (2a)^2 - 4(a^2 - db^2) \in \mathbb{Z}$, and hence that $2b \in \mathbb{Z}$ since d is square-free. Writing $a = m/2$ and $b = n/2$ with $m, n \in \mathbb{Z}$, the condition $a^2 - db^2 = \frac{m^2 - dn^2}{4} \in \mathbb{Z}$ means that:

- m and n have the same parity if $d \equiv 1 \pmod{4}$,
- m and n are both even if $d \equiv 2, 3 \pmod{4}$.

We deduce that:

$$\mathcal{O}_L = \begin{cases} \{a + b\sqrt{d} : (a, b) \in \mathbb{Z}^2 \cup (1/2 + \mathbb{Z})^2\} = \mathbb{Z}[\frac{1+\sqrt{d}}{2}] & \text{if } d \equiv 1 \pmod{4}, \\ \{a + b\sqrt{d} : (a, b) \in \mathbb{Z}^2\} = \mathbb{Z}[\sqrt{d}] & \text{if } d \equiv 2, 3 \pmod{4}. \end{cases}$$

$\square$

In order to study other more complicated rings of integers and to develop a theoretical method to compute them, we need to introduce a certain number of algebraic tools. Given an element $x \in L$, Proposition 1.1.4 provides a simple way to find a non-zero polynomial $f \in K[T]$ such that $f(x) = 0$. Indeed, the multiplication-by- x map:

$$\begin{aligned} m_x : L &\rightarrow L \\ y &\mapsto xy \end{aligned}$$

is a K -linear map and x is a root of its characteristic polynomial:

$$\chi_{L/K, x}(T) := \det(T\text{Id}_L - m_x).$$

Definition 1.1.16

The polynomial $\chi_{L/K,x}(T) := \det(T\text{Id}_L - m_x)$ is called the **characteristic polynomial of x** . One also defines the **trace** and the **norm** of x as:

$$\text{Tr}_{L/K}(x) := \text{Tr}(m_x), \quad N_{L/K}(x) := \det(m_x).$$

If we write $\chi_{L/K,x}(T) = T^n - c_1 T^{n-1} + \dots + (-1)^n c_n$, we then have:

$$n = [L : K], \quad c_1 = \text{Tr}_{L/K}(x), \quad c_n = N_{L/K}(x).$$

The trace defines an K -linear map $\text{Tr}_{L/K} : L \rightarrow K$ and the norm defines a group homomorphism $N_{L/K} : L^\times \rightarrow K^\times$.

Let us now focus on the case where L/K is a separable extension and fix a separable closure L^s of L . According to the Primitive Element Theorem, we can write $L = K(\alpha)$ for some $\alpha \in L$. Denote by $\mu_\alpha \in K[T]$ the minimal polynomial of α . The field L can then be identified with $K[T]/(\mu_\alpha(T))$, and hence the field embeddings of L into L^s fixing K can be described as:

$$\begin{aligned} \iota_\beta : L = K[T]/(\mu_\alpha(T)) &\rightarrow L^s \\ P(T) &\mapsto P(\beta) \end{aligned}$$

for some root β of μ_α . This shows that the set of such embeddings, which we denote $\text{Hom}_K(L, L^s)$, has exactly $n = [L : K] = \deg \mu_\alpha$ elements.

Proposition 1.1.17

With the previous notations, given an element $x \in L$ with minimal polynomial μ_x , we have:

$$\begin{aligned} \chi_{L/K,x}(T) &= \mu_x(T)^{[L:K(x)]} = \prod_{\sigma \in \text{Hom}_K(L, L^s)} (T - \sigma(x)), \\ \text{Tr}_{L/K}(x) &= \sum_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x), \\ N_{L/K}(x) &= \prod_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x). \end{aligned}$$

Proof. Set:

$$u := [K(x) : K] = \deg \mu_x, \quad v := [L : K(x)], \quad n := uv = [L : K],$$

and write:

$$\mu_x(T) = T^u + c_1 T^{u-1} + \dots + c_u \in K[T].$$

The family $(1, x, x^2, \dots, x^{u-1})$ is an K -basis of $K(x)$. Hence, if $(\alpha_1, \dots, \alpha_\nu)$ stands for some $K(x)$ -basis of L , the family:

$$(\alpha_1, \alpha_1 x, \dots, \alpha_1 x^{u-1}, \alpha_2, \alpha_2 x, \dots, \alpha_2 x^{u-1}, \dots, \alpha_\nu, \alpha_\nu x, \dots, \alpha_\nu x^{u-1})$$

is a K -basis of L . In that K -basis, the matrix of the multiplication-by- x map m_x is a block diagonal matrix, that has exactly ν diagonal blocks all of them equal to:

$$C(\mu_x) := \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & -c_u \\ 1 & 0 & 0 & \dots & 0 & -c_{u-1} \\ 0 & 1 & 0 & \dots & 0 & -c_{u-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & -c_{u-1} \end{pmatrix}.$$

Since $C(\mu_x)$ has characteristic polynomial μ_x , we deduce that $\chi_{L/K,x}(T) = \mu_x(T)^\nu$. But $\mu_x(T) = \prod_{\sigma \in \text{Hom}_K(K(x), L^s)} (T - \sigma(x))$, and hence:

$$\chi_{L/K,x}(T) = \prod_{\sigma \in \text{Hom}_K(K(x), L^s)} (T - \sigma(x))^\nu.$$

To conclude, it only remains to check that, for each $\sigma \in \text{Hom}_K(K(x), L^s)$, there exist exactly ν embeddings $\tilde{\sigma} \in \text{Hom}_K(L, L^s)$ that extend σ . To do so, one can apply the Primitive Element Theorem to write $L = K(x, y)$ for some $y \in L$, so that $L \cong K(x)[T]/(\mu_y(T))$ with μ_y the minimal polynomial of y over $K(x)$. The embeddings $\tilde{\sigma} \in \text{Hom}_K(L, L^s)$ that extend σ can then be described as:

$$\begin{aligned} \iota_{\sigma, \beta} : L = K(x)[T]/(\mu_y(T)) &\rightarrow L^s \\ P(T) &\mapsto \sigma(P)(\sigma(\beta)) \end{aligned}$$

for some root β of μ_y . This shows that there are exactly $\nu = [L : K(x)] = \deg \mu_y$ such embeddings. $\square$

Corollary 1.1.18

An element $x \in L$ belongs to B if, and only if, the coefficients of $\chi_{L/K,x}$ belong to A . In particular, if $x \in B$, then $\text{Tr}_{L/K}(x)$ and $N_{L/K}(x)$ both belong to A .

Proof. Assume that x is integral over A . Then $\sigma(x)$ is also integral over A for every $\sigma \in \text{Hom}_K(L, L^s)$. Hence, according to Proposition 1.1.17, all coefficients of $\chi_{L/K,x}$ are integral over A and belong to K . Since A is integrally closed, they belong to A . The converse and the statement about the trace and the norm are obvious. $\square$

Let now $\mathbf{x} := (x_1, \dots, x_n)$ be a K -basis of L . Take some $x \in L$ and write $x = \sum_{i=1}^n \lambda_i x_i$ for some $\lambda_1, \dots, \lambda_n \in K$. We want to find some necessary conditions for x to be in B in terms of the coordinates $\lambda_1, \dots, \lambda_n$.

For that purpose, we start by reducing to the case where the K -basis $\mathbf{x}$ is contained in B thanks to the following Lemma:

Lemma 1.1.19. *The ring B has fraction field L .*

Proof. Take $y \in L$, let $\mu_y = T^d + a_{d-1}T^{d-1} + \dots + a_0 \in K[T]$ be the minimal polynomial y , and write $a_i = b_i/c_i$ with $b_i, c_i \in A$ and $c_i \neq 0$ for each i . If we set $c = c_0 \dots c_{d-1}$, then cy is a root of the polynomial $T^d + a_{d-1}cT^{d-1} + \dots + a_0c^d$, which is monic and has coefficients in A . We deduce that cy is in B , and hence y is in the fraction field of B . $\square$

Up to multiplying the K -basis $\mathbf{x}$ by a suitable element of A , we henceforth assume that $\mathbf{x}$ lies in B . Now, by Corollary 1.1.18.3, for $x = \sum_{i=1}^n \lambda_i x_i$ to be in BK , it is necessary that, for each $1 \leq j \leq n$:

$$\mathrm{Tr}_{L/K}(xx_j) = \sum_{i=1}^n \lambda_i \mathrm{Tr}_{L/K}(x_i x_j) \in A.$$

Setting $M_{\mathbf{x}} := (\mathrm{Tr}_{L/K}(x_i x_j))_{1 \leq i, j \leq n} \in \mathcal{M}_n(A)$, this system of equations can be rewritten as:

$$M_{\mathbf{x}} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \in A^n.$$

Hence:

$$\det(M_{\mathbf{x}}) \cdot \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = {}^t\mathrm{Com}(M_{\mathbf{x}}) \cdot M_{\mathbf{x}} \cdot \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \in A^n,$$

and for each i , the rational number $\det(M_{\mathbf{x}})\lambda_i$ is in A . In other words, we have proved the following proposition:

Proposition 1.1.20

Let $\mathbf{x} = (x_1, \dots, x_n)$ be a K -basis of L that is contained in B . Then:

$$d(\mathbf{x}) \cdot B \subset \bigoplus_{i=1}^n A \cdot x_i,$$

where:

$$d(\mathbf{x}) := \det(\mathrm{Tr}_{L/K}(x_i x_j))_{1 \leq i, j \leq n}.$$

The element $d(\mathbf{x})$ is called the **discriminant** of the basis $\mathbf{x}$. According to Proposition 1.1.17, if $\sigma_1, \dots, \sigma_n$ stand for the field embeddings of L in L^s fixing K , we have:

$$(\mathrm{Tr}_{L/K}(x_i x_j))_{1 \leq i, j \leq n} = \left(\sum_{r=1}^n \sigma_r(x_i) \sigma_r(x_j) \right)_{1 \leq i, j \leq n} = {}^t (\sigma_i(x_j))_{1 \leq i, j \leq n} \cdot (\sigma_i(x_j))_{1 \leq i, j \leq n},$$

and hence the discriminant can be computed as:

$$d(\mathbf{x}) := \det(\mathrm{Tr}_{L/K}(x_i x_j))_{1 \leq i, j \leq n} = \left(\det(\sigma_i(x_j))_{1 \leq i, j \leq n} \right)^2.$$

A particularly interesting case is that of a basis $\mathbf{x}$ of the form $(1, x, x^2, \dots, x^{n-1})$. In that case, we get:

$$d(1, x, x^2, \dots, x^{n-1}) = \left(\det(\sigma_i(x)^{j-1})_{1 \leq i, j \leq n} \right)^2,$$

and hence, by recognizing a Vandermonde determinant:

$$d(1, x, x^2, \dots, x^{n-1}) = \left(\prod_{i < j} (\sigma_i(x) - \sigma_j(x)) \right)^2.$$

If $\mu_x = \prod_{i=1}^n (T - \sigma_i(x))$ is the minimal polynomial of x , this formula can be rewritten as:

$$d(1, x, x^2, \dots, x^{n-1}) = (-1)^{n(n-1)/2} \cdot \prod_{i=1}^n \mu'_x(\sigma_i(x)). \quad (1.1)$$

Denoting by $\alpha_1, \dots, \alpha_{n-1}$ the roots of μ'_x , we get:

$$\begin{aligned} d(1, x, x^2, \dots, x^{n-1}) &= (-1)^{n(n-1)/2} \cdot n^n \cdot \prod_{i=1}^n \prod_{j=1}^{n-1} (\sigma_i(x) - \alpha_j) \\ &= (-1)^{n(n-1)/2} \cdot n^n \cdot \prod_{j=1}^{n-1} \mu_x(\alpha_j). \end{aligned} \quad (1.2)$$

Equations (1.1) and (1.2) are often useful to compute $d(1, x, x^2, \dots, x^{n-1})$.

Coming back to Proposition 1.1.20, for its statement to be not empty, the discriminant $d(\mathbf{x})$ has to be non-zero. This is in fact always the case, as the following proposition shows:

Proposition 1.1.21

Let $\mathbf{x} = (x_1, \dots, x_n)$ be a K -basis of L . The discriminant $d(\mathbf{x})$ is non-zero and the trace bilinear form:

$$\begin{aligned} b: L \times L &\rightarrow K \\ (x, y) &\mapsto \text{Tr}_{L/K}(xy) \end{aligned}$$

is non-degenerate.

Proof of Proposition 1.1.21. Note that $d(\mathbf{x})$ is the determinant of the trace bilinear form b in the basis $\mathbf{x}$. Hence, to prove the proposition, it suffices to prove that $d(\mathbf{y}) \neq 0$ for *some* K -basis $\mathbf{y}$ of L . To do so, use the Primitive Element Theorem to write $L = K(x)$ for some $y \in L$. The family $\mathbf{y} = (1, y, y^2, \dots, y^{n-1})$ is then a K -basis of L and as we have already seen:

$$d(1, y, y^2, \dots, y^{n-1}) = \left(\prod_{i < j} (\sigma_i(y) - \sigma_j(y)) \right)^2,$$

where $\sigma_1, \dots, \sigma_n$ are the field embeddings of L in L^s fixing K . We conclude by observing that the $\sigma_i(y)$'s are pairwise distinct. $\square$

Assume that $K = \mathbb{Q}$ and $A = \mathbb{Z}$. Now that we know that $d(\mathbf{x})$ is non-zero for every $\mathbb{Q}$ -basis $\mathbf{x}$ of L , Proposition 1.1.20 provides a theoretical way to compute the ring of integers $\mathcal{O}_L$ of the number field L . Here is the step-by-step procedure:

1. Find a $\mathbb{Q}$ -basis $\mathbf{x} = (x_1, \dots, x_n)$ of L that is contained in $\mathcal{O}_L$. For instance, find an element $x \in \mathcal{O}_L$ such that $L = \mathbb{Q}(x)$ and set $\mathbf{x} := (1, x, x^2, \dots, x^{n-1})$.
2. Compute $d(\mathbf{x})$ and deduce that:

$$\bigoplus_{i=1}^n \mathbb{Z} \cdot x_i \subset \mathcal{O}_L \subset \frac{1}{d(\mathbf{x})} \cdot \bigoplus_{i=1}^n \mathbb{Z} \cdot x_i.$$

3. Find representatives $\alpha_1, \dots, \alpha_r \in \frac{1}{d(\mathbf{x})} \cdot \bigoplus_{i=1}^n \mathbb{Z} \cdot x_i$ of the finite quotient:

$$\left(\frac{1}{d(\mathbf{x})} \cdot \bigoplus_{i=1}^n \mathbb{Z} \cdot x_i \right) / \left(\bigoplus_{i=1}^n \mathbb{Z} \cdot x_i \right).$$

4. For each i , compute the characteristic polynomial $\chi_{L/\mathbb{Q}, \alpha_i}$ of α_i and determine whether it has integer coefficients.

5. Deduce that:

$$\mathcal{O}_L = \bigsqcup_{\substack{i=1 \\ \chi_L/\mathbb{Q}, \alpha_i \in \mathbb{Z}[T]}}^r \alpha_i + \left(\bigoplus_{i=1}^n \mathbb{Z} \cdot x_i \right).$$

Example 1.1.22. This method is often more theoretical than practical, because the integer r appearing in step 3 is often quite large. The following two examples illustrate this phenomenon.

- (i) Consider the number field $L = \mathbb{Q}(x)$ with $x \in \mathbb{C}$ such that $x^3 + x + 1 = 0$. Introduce the $\mathbb{Q}$ -basis $\mathbf{x} := (1, x, x^2)$ of L . Since the minimal polynomial of x is $\mu_x = T^3 + T + 1$, the roots of μ'_x are $\pm \frac{i}{\sqrt{3}}$ and the discriminant $d(\mathbf{x})$ can be computed as follows:

$$d(\mathbf{x}) = -3^3 \cdot \mu_x\left(\frac{i}{\sqrt{3}}\right) \cdot \mu_x\left(-\frac{i}{\sqrt{3}}\right) = -27 \left(1 + \frac{2}{3\sqrt{3}}i\right) \left(1 - \frac{2}{3\sqrt{3}}i\right) = -31.$$

We deduce that:

$$\mathbb{Z} \oplus \mathbb{Z}x \oplus \mathbb{Z}x^2 \subset \mathcal{O}_L \subset \frac{1}{31} \cdot (\mathbb{Z} \oplus \mathbb{Z}x \oplus \mathbb{Z}x^2).$$

As representatives of the quotient:

$$\left(\frac{1}{31} \cdot (\mathbb{Z} \oplus \mathbb{Z}x \oplus \mathbb{Z}x^2) \right) / (\mathbb{Z} \oplus \mathbb{Z}x \oplus \mathbb{Z}x^2),$$

one can choose the $\alpha_{a,b,c} = \frac{a+bx+cx^2}{31}$ with $a, b, c \in \{0, 1, \dots, 30\}$. For each such $\alpha_{a,b,c}$ we have to compute its characteristic polynomial and check whether it has integer coefficients. There are therefore $31^3 = 29791$ characteristic polynomials to compute!

- (ii) Similarly, consider the number field $L = \mathbb{Q}(\sqrt[3]{2})$, choose $x = \sqrt[3]{2}$ and introduce the $\mathbb{Q}$ -basis $\mathbf{x} := (1, \sqrt[3]{2}, \sqrt[3]{4})$ of L . Since the minimal polynomial of x is $\mu_x = T^3 - 2$, the discriminant $d(\mathbf{x})$ can be computed as follows:

$$d(\mathbf{x}) = \mu'_x(\sqrt[3]{2}) \cdot \mu'_x(e^{2i\pi/3}\sqrt[3]{2}) \cdot \mu'_x(e^{4i\pi/3}\sqrt[3]{2}) = 2^2 \times 3^3 = 108.$$

We deduce that:

$$\mathbb{Z} \oplus \mathbb{Z}\sqrt[3]{2} \oplus \mathbb{Z}\sqrt[3]{4} \subset \mathcal{O}_L \subset \frac{1}{108} \cdot (\mathbb{Z} \oplus \mathbb{Z}\sqrt[3]{2} \oplus \mathbb{Z}\sqrt[3]{4}).$$

As representatives of the quotient:

$$\left(\frac{1}{108} \cdot (\mathbb{Z} \oplus \mathbb{Z}\sqrt[3]{2} \oplus \mathbb{Z}\sqrt[3]{4}) \right) / (\mathbb{Z} \oplus \mathbb{Z}\sqrt[3]{2} \oplus \mathbb{Z}\sqrt[3]{4}),$$

one can choose the $\alpha_{a,b,c} = \frac{a+b\sqrt[3]{2}+c\sqrt[3]{4}}{108}$ with $a, b, c \in \{0, 1, \dots, 107\}$. There are therefore $108^3 = 1259712$ characteristic polynomials to compute!

In order to better describe rings of integers and reduce the number of characteristic polynomials one needs to check in the previous procedure, one can strengthen Proposition 1.1.20 in the case $K = \mathbb{Q}$ and $A = \mathbb{Z}$.

Proposition 1.1.23

1. Assume that A is a principal ideal domain. The field L has a K -basis $\mathbf{y} = (y_1, \dots, y_n)$ such that:

$$B = Ay_1 \oplus \dots \oplus Ay_n.$$

Such a basis is called an **integral basis of B** .

2. Assume that $A = \mathbb{Z}$ and $K = \mathbb{Q}$. The discriminant $\Delta_L := d(\mathbf{y})$ of an integral basis $\mathbf{y}$ of $\mathcal{O}_L$ is independent of the choice of $\mathbf{y}$ and for every K -basis $\mathbf{x}$ of L contained in $\mathcal{O}_L$:

$$d(\mathbf{x}) = [\mathcal{O}_L : \Gamma_{\mathbf{x}}]^2 \cdot \Delta_L,$$

where $\Gamma_{\mathbf{x}} := \mathbb{Z}x_1 \oplus \dots \oplus \mathbb{Z}x_n$. The discriminant Δ_L is called the **discriminant of L** .

In particular, when $A = \mathbb{Z}$ and $K = \mathbb{Q}$, if $\mathbf{x}$ is a $\mathbb{Q}$ -basis of L contained in $\mathcal{O}_L$ such that $d(\mathbf{x})$ is square-free:

$$B = \bigoplus_{i=1}^n A \cdot x_i.$$

Proof. 1. Take a K -basis $\mathbf{x} = (x_1, \dots, x_n)$ of L that is contained in B . By Proposition 1.1.20:

$$\bigoplus_{i=1}^n A \cdot x_i \subset B \subset \frac{1}{d(\mathbf{x})} \cdot \bigoplus_{i=1}^n A \cdot x_i.$$

By the structure theorem of finitely generated abelian groups, B is a free abelian group of rank n . In other words, we can find $y_1, \dots, y_n \in B$ such that:

$$B = Ay_1 \oplus \dots \oplus Ay_n.$$

The family $\mathbf{y} = (y_1, \dots, y_n)$ is necessarily a K -basis of L .

2. Let $\mathbf{y}$ be an integral basis of $\mathcal{O}_L$ and let $\mathbf{x}$ be any $\mathbb{Q}$ -basis of L contained in $\mathcal{O}_L$. Recall that $d(\mathbf{x})$ (resp. $d(\mathbf{y})$) is the determinant of the trace bilinear form in the basis $\mathbf{x}$ (resp. $\mathbf{y}$). Hence, if P is the transition matrix from $\mathbf{y}$ to $\mathbf{x}$, we have $d(\mathbf{x}) = d(\mathbf{y}) \cdot (\det P)^2$. But by the Smith normal form theorem, $|\det P| = [\mathcal{O}_L : \Gamma_{\mathbf{x}}]$, hence:

$$d(\mathbf{x}) = d(\mathbf{y}) \cdot [\mathcal{O}_L : \Gamma_{\mathbf{x}}]^2.$$

When $\mathbf{x}$ is an integral basis of $\mathcal{O}_L$, we have $\Gamma_{\mathbf{x}} = \mathcal{O}_L$ and hence $d(\mathbf{x}) = d(\mathbf{y})$. $\square$

Example 1.1.24. Let us come-back to examples 1.1.22.

- (i) When $L = \mathbb{Q}(x)$ with $x^3 + x + 1 = 0$, the discriminant $d(1, x, x^2)$ is equal to -31 , and hence is square-free. Hence:

$$\mathcal{O}_L = \mathbb{Z} \oplus \mathbb{Z}x \oplus \mathbb{Z}x^2.$$

- (ii) The case of the number field $L = \mathbb{Q}(\sqrt[3]{2})$ is more complicated. Indeed, this time, if we set $x := \sqrt[3]{2}$, we have $d(1, x, x^2) = 108$, and hence:

$$[\mathcal{O}_L : \mathbb{Z}[\sqrt[3]{2}]] \mid 6.$$

In particular,

$$\mathcal{O}_L \subset \frac{1}{6}\mathbb{Z}[\sqrt[3]{2}],$$

and hence we only need to compute $6^3 = 216$ characteristic polynomials. This is still a lot. In the exercise sessions, we will develop a method based on Eisenstein polynomials that allows to prove that $\mathcal{O}_L = \mathbb{Z}[\sqrt[3]{2}]$.

Remark 1.1.25. The statement of Proposition 1.1.23 can be immediately generalized to the situation where A is a principal ideal domain instead of $\mathbb{Z}$.

1.2 The algebraic behaviour of rings of integers

In order to specify some of the most important algebraic properties of rings of integers, we introduce and study the notion of Dedekind domain.

1.2.1 Dedekind domains

Definition 1.2.1

A domain A is said to be a **Dedekind domain** if it is noetherian, integrally closed and every non-zero prime ideal in A is maximal.

Example 1.2.2. Principal ideal domains are Dedekind domains. However, unique factorization domain do not need to be Dedekind domains (for instance, the ideal (X) in $\mathbb{C}[X, Y]$ is prime but not maximal).

Theorem 1.2.3

Consider the framework 1.1.13 and assume that A is a Dedekind domain. Then B is also a Dedekind domain. In particular, rings of integers of number fields are Dedekind domains.

Proof. We already know that B is integrally closed by Remark 1.1.12. Moreover, if $\mathbf{x}$ is a K -basis of L contained in B , Proposition 1.1.20 shows that:

$$B \subset \frac{1}{d(\mathbf{x})} \bigoplus_{i=1}^n Ax_i.$$

Hence B is a sub- A -module of a finitely generated A -module. Since A is a noetherian ring, we deduce that B is a noetherian A -module. In particular, it is a noetherian ring.

It remains to show that every non-zero prime ideal in B is maximal. Let $\mathfrak{q}$ be such a prime ideal and consider $\mathfrak{p} = \mathfrak{q} \cap A$. It is a prime ideal of A . Moreover, if x is a non-zero element of $\mathfrak{q}$ and we let:

$$\mu = X^d + a_{d-1}X^{d-1} + \cdots + a_0 \in A[X]$$

be the minimal polynomial of x , then:

$$a_0 = -\left(x^d + a_{d-1}x^{d-1} + \cdots + a_1x\right) \in (\mathfrak{q} \cap A) \setminus \{0\} = \mathfrak{p} \setminus \{0\}.$$

We deduce that $\mathfrak{p} \neq 0$, and hence $\mathfrak{p}$ is a maximal ideal of A . In particular, $B/\mathfrak{q}$ is a domain that is, at the same time, a finite-dimensional $(A/\mathfrak{p})$ -vector space. It is therefore a field. $\square$



Richard Dedekind (Braunschweig 1831 - Braunschweig 1916)

1.2.2 Fractional ideals

In general, Dedekind domains are not principal ideal domains nor unique factorization domains, and hence their arithmetic behaviour is more complicated. However, they have a very nice property: even if elements do not factor uniquely as products of irreducible elements, ideals do factor uniquely as products of prime ideals. To see this, we first study the notion of fractional ideals.

Definition 1.2.4

Let A be a Dedekind domain with fraction field K and let I be a subset of K . We say that I is a **fractional ideal** of A if there exists an element $a \in A \setminus \{0\}$ such that aI is an ideal of A .

Example 1.2.5. If A is a principal ideal domain, its fractional ideals are the subsets of K of the form xA with $x \in K$. In particular, if A is not a field, one can check that K itself is not a fractional ideal.

As for usual ideals, one can define several arithmetical operations on fractional ideals. Indeed, if I and J are two fractional ideals of A , then:

$$I \cap J, \quad I + J := \{i + j : i \in I, j \in J\}, \quad IJ := \left\{ \sum_{r=1}^n i_r j_r : n \geq 0, i_1, \dots, i_n \in I, j_1, \dots, j_n \in J \right\}$$

are all fractional ideals. A fractional ideal I is said to be **invertible** if there exists a fractional ideal J such that $IJ = A$.

Theorem 1.2.6

Let A be a Dedekind domain and let $\mathfrak{p}$ be a non-zero prime ideal in A . Then $\mathfrak{p}$ is an invertible fractional ideal.

Observe that, if J is a fractional ideal such that $\mathfrak{p}J = A$, then

$$J \subset \{x \in K : x\mathfrak{p} \subset A\}.$$

We therefore naturally define:

$$\mathfrak{p}^{-1} := \{x \in K : x\mathfrak{p} \subset A\}.$$

If y is a non-zero element in $\mathfrak{p}$, then $y\mathfrak{p}^{-1}$ is an ideal in A . Hence $\mathfrak{p}^{-1}$ is a fractional ideal in A . Theorem 1.2.6 is then an immediate consequence of the following proposition:

Proposition 1.2.7

Let A be a Dedekind domain, let $\mathfrak{p}$ be a prime ideal in A and let I be any non-zero ideal in A . Then $\mathfrak{p}^{-1}I$ strictly contains I . In particular, $\mathfrak{p}^{-1}\mathfrak{p} = A$.

Proof. We carry out the proof in three main steps.

Step 1: We start by proving that every non-zero ideal in A contains a product of prime ideals. For that purpose, consider the family $\mathcal{E}$ of non-zero ideals in A that do not satisfy this property and assume by contradiction that $\mathcal{E}$ is non-empty. Since A is noetherian, the family $\mathcal{E}$ contains a maximal element I_0 . Since $I_0 \in \mathcal{E}$, the ideal I_0 cannot be prime. This means that there exist $a, b \in A \setminus I_0$ such that $ab \in I_0$. By maximality of I_0 , the ideals $I_0 + (a)$ and $I_0 + (b)$ both contain a product of prime ideals. But then so does $(I_0 + (a))(I_0 + (b)) \subset I_0$. This contradicts that $I_0 \in \mathcal{E}$.

Step 2: We now prove that $\mathfrak{p}^{-1}$ strictly contains A . By definition, $\mathfrak{p}^{-1}$ contains A . It remains to construct an element that lies in $\mathfrak{p}^{-1}$ but not in A . To do so, take a non-zero element $a \in \mathfrak{p}$ and apply Step 1 to find prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ such that :

$$\mathfrak{p}_1 \dots \mathfrak{p}_n \subset (a) \subset \mathfrak{p}. \quad (1.3)$$

Assume without loss of generality that n is as small as possible. By inclusion (1.3) and the primality of the ideal $\mathfrak{p}$, one of the $\mathfrak{p}_i$'s is contained in $\mathfrak{p}$. Assume without loss of generality that $\mathfrak{p}_n \subset \mathfrak{p}$. By maximality of $\mathfrak{p}_n$, we deduce that $\mathfrak{p}_n = \mathfrak{p}$. Moreover, by minimality of n , we can find an element $b \in \mathfrak{p}_1 \dots \mathfrak{p}_{n-1}$ that does not lie in (a) . This implies that $a^{-1}b$ belongs to $\mathfrak{p}^{-1}$ but not to A , as wished.

Step 3: We finally prove that the fractional ideal $\mathfrak{p}^{-1}I$ strictly contains I , and in particular that $\mathfrak{p}^{-1}\mathfrak{p} = A$. Assume by contradiction that $\mathfrak{p}^{-1}I = I$ and take $x \in \mathfrak{p}^{-1}$. We have $xI \subset I$, and since A is noetherian, the ideal I is finitely generated. By Proposition 1.1.4, we deduce that x is a root of a monic polynomial with coefficients in A . But A is integrally closed, hence $x \in A$ and $\mathfrak{p}^{-1} = A$. This contradicts Step 2, and therefore shows that $\mathfrak{p}^{-1}I$ strictly contains I . In particular, when $I = \mathfrak{p}$, we obtain $\mathfrak{p}\mathfrak{p}^{-1}$ is an ideal of A that strictly contains $\mathfrak{p}$. By maximality of $\mathfrak{p}$, we conclude that $\mathfrak{p}\mathfrak{p}^{-1} = A$. $\square$

We will prove in the next section that in fact every non-zero fractional ideal in a Dedekind domain is invertible. To do so, we will first prove that, in that context, every ideal is a product of prime ideals

1.2.3 Factorization of ideals

We are now ready to prove the unique factorization of ideals as products of prime ideals in Dedekind domains.

Theorem 1.2.8

Let A be a Dedekind domain and let I be a non-zero ideal in A . Then there exist an integer $n \geq 0$ and some non-zero prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ in A such that $I = \mathfrak{p}_1 \dots \mathfrak{p}_n$. The integer n is uniquely determined, and the prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ are uniquely determined up to the order.

Proof. We start by proving the existence. Let $\mathcal{E}$ be the family of ideals in A other than (0) and (1) that are not a product of prime ideals. By contradiction, assume that $\mathcal{E}$ is non-empty. Since A is noetherian, the family $\mathcal{E}$ contains a maximal element I_0 . By Krull's Theorem, I_0 is contained in a maximal ideal $\mathfrak{p}$. By Proposition 1.2.7, the ideal I_0 is strictly contained in $\mathfrak{p}^{-1}I_0$, and therefore $\mathfrak{p}^{-1}I_0$ is a product of prime ideals. Hence by using again Proposition 1.2.7, the ideal $I_0 = \mathfrak{p}\mathfrak{p}^{-1}I_0$ is also a product of prime ideals. This contradicts that $I_0 \in \mathcal{E}$.

We now move on to the uniqueness. Consider non-zero prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_m, \mathfrak{q}_1, \dots, \mathfrak{q}_n$ such that:

$$\mathfrak{p}_1 \dots \mathfrak{p}_m = \mathfrak{q}_1 \dots \mathfrak{q}_n.$$

We prove by induction on m that $m = n$ and that the $\mathfrak{p}_i$'s and the $\mathfrak{q}_j$'s coincide up to the order. Of course, the result is obvious for $m = 0$. We henceforth assume that $m > 0$. Since $\mathfrak{p}_1 \dots \mathfrak{p}_m \subset \mathfrak{q}_n$ and $\mathfrak{q}_n$ is prime, one of the ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ is contained in $\mathfrak{q}_n$. Without loss of generality, assume that $\mathfrak{p}_m \subset \mathfrak{q}_n$. Since A is a Dedekind domain, $\mathfrak{p}_m$ is a maximal ideal in A , and hence $\mathfrak{p}_m = \mathfrak{q}_n$. Therefore, by Proposition 1.2.7:

$$\mathfrak{p}_1 \dots \mathfrak{p}_{m-1} = \mathfrak{p}_1 \dots \mathfrak{p}_m \mathfrak{p}_m^{-1} = \mathfrak{q}_1 \dots \mathfrak{q}_n \mathfrak{q}_n^{-1} = \mathfrak{q}_1 \dots \mathfrak{q}_{n-1}.$$

By our inductive assumption, $m-1 = n-1$ and the $\mathfrak{p}_i$'s for $i \leq m-1$ and the $\mathfrak{q}_j$'s for $j \leq m-1$ coincide up to the order. We conclude that $m = n$ and that the $\mathfrak{p}_i$'s for $i \leq m$ and the $\mathfrak{q}_j$'s for $j \leq n$ coincide up to the order. $\square$

Corollary 1.2.9

Every non-zero fractional ideal I in a Dedekind domain is invertible and can be uniquely written as:

$$I = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\alpha_{\mathfrak{p}}}$$

for some finitely supported family $(\alpha_{\mathfrak{p}})_{\mathfrak{p}}$ with values in $\mathbb{Z}$.

Proof. Let I be a non-zero fractional ideal of a Dedekind domain A . There exists $x \in A \setminus \{0\}$ such that xI is an ideal in A . By Theorem 1.2.8, we can write:

$$xI = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\beta_{\mathfrak{p}}}, \quad (x) = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\gamma_{\mathfrak{p}}}$$

for some finitely supported families $(\beta_{\mathfrak{p}})_{\mathfrak{p}}$ and $(\gamma_{\mathfrak{p}})_{\mathfrak{p}}$ with values in $\mathbb{Z}_{\geq 0}$. Then $I = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\beta_{\mathfrak{p}} - \gamma_{\mathfrak{p}}}$ and I is invertible with inverse $I^{-1} = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\gamma_{\mathfrak{p}} - \beta_{\mathfrak{p}}}$.

As for the uniqueness statement, assume:

$$I = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\alpha_{\mathfrak{p}}} = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\alpha'_{\mathfrak{p}}}$$

for some finitely supported families $(\alpha_{\mathfrak{p}})_{\mathfrak{p}}$ and $(\alpha'_{\mathfrak{p}})_{\mathfrak{p}}$. If m is an integer greater than all the $-\alpha_{\mathfrak{p}}$ and all the $-\alpha'_{\mathfrak{p}}$, then:

$$\prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\alpha_{\mathfrak{p}} + m} = \prod_{\mathfrak{p} \text{ prime}} \mathfrak{p}^{\alpha'_{\mathfrak{p}} + m}.$$

Hence $\alpha_{\mathfrak{p}} + m = \alpha'_{\mathfrak{p}} + m$ and $\alpha_{\mathfrak{p}} = \alpha'_{\mathfrak{p}}$ for every $\mathfrak{p}$. □

Remark 1.2.10. The previous corollary is telling that the set $\mathcal{I}(K)$ of non-zero fractional ideals in a Dedekind domain is a free abelian group if endowed with the multiplication of fractional ideals.

Corollary 1.2.11

Let I and J be two non-zero ideals in a Dedekind domain A . Then $I \subseteq J$ if, and only if, J divides I .

Proof. Assume that $I \subseteq J$ and set $I' := J^{-1}I$. Then $I' \subseteq J^{-1}J = \mathcal{O}_K$ and $I = I'J$. The converse is obvious. □

1.2.4 $\mathfrak{p}$ -adic valuations

Definition 1.2.12

Let A be a Dedekind domain with fraction field K . Let $\mathfrak{p}$ be a non-zero prime ideal in A . We define a map $\nu_{\mathfrak{p}} : K \rightarrow \mathbb{Z} \cup \{\infty\}$, called the **$\mathfrak{p}$ -adic valuation**, in the following way: we set $\nu_{\mathfrak{p}}(0) = \infty$ and for $x \in K^\times$, we let $\nu_{\mathfrak{p}}(x)$ be the power of $\mathfrak{p}$ in the prime factorization of the fractional ideal (x) .

The $\mathfrak{p}$ -adic valuation satisfies some nice properties:

- $\forall x \in K, \nu_{\mathfrak{p}}(x) = 0 \Leftrightarrow x = 0$;
- $\forall x, y \in K, \nu_{\mathfrak{p}}(xy) = \nu_{\mathfrak{p}}(x) + \nu_{\mathfrak{p}}(y)$;
- $\forall x, y \in K, \nu_{\mathfrak{p}}(x + y) \geq \min\{\nu_{\mathfrak{p}}(x), \nu_{\mathfrak{p}}(y)\}$, with equality when $\nu_{\mathfrak{p}}(x) \neq \nu_{\mathfrak{p}}(y)$.

Moreover, the Dedekind ring A can be characterised inside its fraction field thanks to the valuations:

$$A = \{x \in K : \forall \mathfrak{p} \text{ prime}, \nu_{\mathfrak{p}}(x) \geq 0\}.$$

Proposition 1.2.13

Let A be a Dedekind domain with fraction field K . Let $\mathfrak{p}$ be a non-zero prime ideal in A . The set $\nu_{\mathfrak{p}}^{-1}(\mathbb{Z}_{\geq 0})$ coincides with the localization $A_{\mathfrak{p}}$ of A at $\mathfrak{p}$. It is a local principal ideal domain with residue field $k(\mathfrak{p}) = A/\mathfrak{p}$.

Proof. Given $x = y/z \in A_{\mathfrak{p}}$ with $y \in A$ and $z \in A \setminus \mathfrak{p}$, we have $\nu_{\mathfrak{p}}(x) = \nu_{\mathfrak{p}}(y) - \nu_{\mathfrak{p}}(z) = \nu_{\mathfrak{p}}(y) \geq 0$, hence $x \in \nu_{\mathfrak{p}}^{-1}(\mathbb{Z}_{\geq 0})$. Conversely, take $x = y/z \in \nu_{\mathfrak{p}}^{-1}(\mathbb{Z}_{\geq 0})$, with $y, z \in A$. Set $m = \nu_{\mathfrak{p}}(y)$ and $n = \nu_{\mathfrak{p}}(z)$, so that $m \geq n$. Take $t \in \mathfrak{p}^{-n} \setminus \mathfrak{p}^{-n+1}$. Then $x = (yt^{-1})/(zt^{-1})$ and yt^{-1} and zt^{-1} belong to $\bigcap_{\mathfrak{p}} \nu_{\mathfrak{p}}^{-1}(\mathbb{Z}_{\geq 0}) = A$. Moreover $\nu_{\mathfrak{p}}(zt^{-1}) = 0$, so $zt^{-1} \notin \mathfrak{p}$. Hence $x \in A_{\mathfrak{p}}$, and $\nu_{\mathfrak{p}}^{-1}(\mathbb{Z}_{\geq 0}) = A_{\mathfrak{p}}$.

By the usual properties of localization at a prime ideal, the unique maximal ideal of $A_{\mathfrak{p}}$ is $\mathfrak{p}A_{\mathfrak{p}}$, so $A_{\mathfrak{p}}$ is local, and the residue field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is canonically isomorphic to $\text{Frac}(A/\mathfrak{p}) = A/\mathfrak{p}$. Now let I be a non-zero ideal of $A_{\mathfrak{p}}$, and let $x_0 \in I$ be an element with minimal $\mathfrak{p}$ -adic valuation. Given $x \in I$, we have $\nu_{\mathfrak{p}}(x/x_0) \geq 0$, hence $x/x_0 \in A_{\mathfrak{p}}$. We deduce that x_0 divides x in $A_{\mathfrak{p}}$, so that $I = x_0A_{\mathfrak{p}}$. We conclude that $A_{\mathfrak{p}}$ is a principal ideal domain. $\square$

Remark 1.2.14. The previous proposition is very convenient to work with Dedekind domains. It often allows to reduce questions on Dedekind domains to the corresponding questions for principal ideal domains.

1.2.5 Behaviour under finite extensions

In this whole section, we consider the Framework 1.1.13 and we let $\mathfrak{p}$ be a non-zero prime ideal of A . The ideal $\mathfrak{p}B$ of B can then be written as:

$$\mathfrak{p}B = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_g^{e_g}$$

where $e_1, \dots, e_g$ are positive integers and $\mathfrak{p}_1, \dots, \mathfrak{p}_g$ are exactly those prime ideals of B lying over $\mathfrak{p}$ (in the sense that $\mathfrak{p}_i \cap A = \mathfrak{p}$).

Definition 1.2.15

The exponent $e_i = e(\mathfrak{p}_i/\mathfrak{p})$ is called the **ramification index** of $\mathfrak{p}_i$ over $\mathfrak{p}$. The degree $f_i = f(\mathfrak{p}_i/\mathfrak{p}) := [k(\mathfrak{p}_i) : k(\mathfrak{p})]$ is called the **residual degree** of $\mathfrak{p}_i$ over $\mathfrak{p}$. The number of prime ideals $g = g_L(\mathfrak{p})$ is called the **splitting number** of $\mathfrak{p}$.

Remark 1.2.16. Ramification indices and residual degrees behave multiplicatively in the following sense: if L' is a finite extension of L and $\mathfrak{p}'_i$ is some prime ideal in the integral closure B' of A in L' that lies above $\mathfrak{p}_i$, then $e(\mathfrak{p}'_i/\mathfrak{p}) = e(\mathfrak{p}'_i/\mathfrak{p}_i)e(\mathfrak{p}_i/\mathfrak{p})$ and $f(\mathfrak{p}'_i/\mathfrak{p}) = f(\mathfrak{p}'_i/\mathfrak{p}_i)f(\mathfrak{p}_i/\mathfrak{p})$. Indeed, the second equality is obvious. For the first one, we can write $\mathfrak{p}B = \mathfrak{p}_1^{e(\mathfrak{p}_i/\mathfrak{p})} \mathfrak{a}$ for some ideal $\mathfrak{a}$ coprime to $\mathfrak{p}_1$ and $\mathfrak{p}_1 B' = \mathfrak{p}_1^{e(\mathfrak{p}'_i/\mathfrak{p}_i)} \mathfrak{a}'$ for some ideal $\mathfrak{a}'$ coprime to $\mathfrak{p}'_i$, so that $\mathfrak{p}B' = \mathfrak{p}_1^{e(\mathfrak{p}'_i/\mathfrak{p}_i)e(\mathfrak{p}_i/\mathfrak{p})} \mathfrak{a}'^{e(\mathfrak{p}_i/\mathfrak{p})} (\mathfrak{a}B')$ with $\mathfrak{a}'^{e(\mathfrak{p}_i/\mathfrak{p})} (\mathfrak{a}B')$ coprime to $\mathfrak{p}'_i$. Hence $e(\mathfrak{p}'_i/\mathfrak{p}) = e(\mathfrak{p}'_i/\mathfrak{p}_i)e(\mathfrak{p}_i/\mathfrak{p})$.

Proposition 1.2.17

Recall that $\mathfrak{p}B = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_g^{e_g}$. We have the relation:

$$\sum_{i=1}^g e_i f_i = [L : K].$$

Proof. For each $i \neq j$, we have $\mathfrak{p}_i^{e_i} + \mathfrak{p}_j^{e_j} = B$. Hence, by the Chinese remainder Theorem, we have:

$$B/\mathfrak{p}B \cong \bigoplus_i B/\mathfrak{p}_i^{e_i} B.$$

This is an isomorphism of $k(\mathfrak{p})$ -vector spaces, and by comparing dimensions, we see that it suffices to prove that $\dim B/\mathfrak{p}B = [L : K]$ and that $\dim B/\mathfrak{p}_i^{e_i} B = e_i f_i$ for each i .

To show the equality $\dim B/\mathfrak{p}_i^{e_i} B = e_i f_i$, observe that, for each $m \geq 0$ and $x \in \mathfrak{p}_i^m \setminus \mathfrak{p}_i^{m+1}$, the $k(\mathfrak{p}_i)$ -linear map:

$$\begin{aligned} k(\mathfrak{p}_i) &\rightarrow \mathfrak{p}_i^m / \mathfrak{p}_i^{m+1} \\ b \bmod \mathfrak{p}_i &\mapsto bx \bmod \mathfrak{p}_i^{m+1} \end{aligned}$$

is an isomorphism. This means that the quotient $\mathfrak{p}_i^m / \mathfrak{p}_i^{m+1}$ is a one-dimensional $k(\mathfrak{p}_i)$ -vector space and hence an f_i -dimensional $k(\mathfrak{p})$ -vector space. This being true for all m , we deduce that $B/\mathfrak{p}_i^{e_i}$ has dimension $e_i f_i$ over $k(\mathfrak{p})$.

Let us now show that $\dim B/\mathfrak{p}B = [L : K]$. Set $B_{\mathfrak{p}} = BA_{\mathfrak{p}}$. One easily checks that $B_{\mathfrak{p}}$ is the integral closure of $A_{\mathfrak{p}}$ in L . Indeed, if we take $x = b/s \in B_{\mathfrak{p}}$ with $b \in B$ and $s \in A \setminus \mathfrak{p}$ and we let $\mu_b = X^d + a_{d-1}X^{d-1} + \dots + a_0 \in A[X]$ be the minimal polynomial of b , then $\mu_x := \frac{\mu_b(sX)}{s^d}$ is a monic polynomial with coefficients in $A_{\mathfrak{p}}$ that has x as a root. Since $A_{\mathfrak{p}}$ is a principal ideal domain, we deduce that $B_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$ -module of rank $[L : K]$. Hence $\dim_{k(\mathfrak{p})} B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} = [L : K]$.

We conclude the proof by observing that the natural morphism $\varphi : B/\mathfrak{p}B \rightarrow B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ induced by the inclusion $B \subset B_{\mathfrak{p}}$ is an isomorphism. Indeed, if $b \in B \cap \mathfrak{p}B_{\mathfrak{p}}$, we can write $b = \pi/s$ with $\pi \in \mathfrak{p}B$ and $s \in A \setminus \mathfrak{p}$. But $\mathfrak{p}$ being a maximal ideal in A , we can write $st - 1 = \pi'$ with $\pi' \in \mathfrak{p}$ and $t \in A$. And then $b = \pi t - \pi' b \in \mathfrak{p}B$. This shows that φ is injective. As for surjectivity, take some $b/s \in B_{\mathfrak{p}}$ for some $b \in B$ and $s \in A \setminus \mathfrak{p}$. As before, we can write $st - 1 = \pi'$ with $\pi' \in \mathfrak{p}$ and $t \in A$. And then $\varphi(tb) = tb = \frac{b}{s} + \frac{\pi' b}{s} = \frac{b}{s} \in B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$. $\square$

The following proposition allows to compute the prime factorization of $\mathfrak{p}B$ for almost every prime ideal $\mathfrak{p}$ of A :

Proposition 1.2.18

Let $\alpha \in B$ be a primitive element for the extension L/K with minimal polynomial μ and set $B' = A[\alpha]$. Let $\mathfrak{p}$ be a prime ideal in A such that $\mathfrak{p}B$ is coprime to the **conductor ideal**:

$$\mathfrak{f}(B/B') := \{x \in B : xB \subset B'\} \subset B.$$

Write the factorization into irreducibles of the reduction of μ modulo $\mathfrak{p}$:

$$\bar{\mu} = \bar{\mu}_1^{e_1} \dots \bar{\mu}_g^{e_g}.$$

For each $1 \leq i \leq g$, let μ_i be a lifting of $\bar{\mu}_i$ in $A[X]$ and set $\mathfrak{p}_i := \mathfrak{p} + \mu_i(\theta)B \subset B$. Then $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are the different prime ideals lying above $\mathfrak{p}$, their corresponding residual degrees are the degrees of the polynomials $\bar{\mu}_1, \dots, \bar{\mu}_r$, and $\mathfrak{p}B = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_g^{e_g}$.

Proof. We compute the quotient $B/\mathfrak{p}B$. To do so, we start by considering the natural morphism $\varphi : B'/\mathfrak{p}B' \rightarrow B/\mathfrak{p}B$ and we check that it is an isomorphism. Indeed, take some $b' \in B' \cap \mathfrak{p}B$ and write $b' = \sum_i \pi_i b_i$ with $\pi_i \in \mathfrak{p}$ and $b_i \in B$. Since $\mathfrak{p}B$ and $\mathfrak{f}(B/B')$ are coprime, we can find $\pi \in \mathfrak{p}B$ and $c \in \mathfrak{f}(B'/B)$ such that $\pi + c = 1$. By definition of the conductor, we get $b' = \pi b' + \sum_i \pi_i (cb_i) \in \mathfrak{p}B'$ and hence φ is injective. As for the surjectivity, given $b \in B$, we have $\varphi(cb) = b - \pi b = b \in B/\mathfrak{p}B$.

Observing that $B' \cong A[X]/(\mu)$, we get an isomorphism:

$$\psi : B/\mathfrak{p}B \xrightarrow[\cong]{\varphi^{-1}} B'/\mathfrak{p}B' \cong k(\mathfrak{p})[X]/(\bar{\mu}).$$

The prime ideals of B containing $\mathfrak{p}B$ therefore correspond to the prime ideals of $k(\mathfrak{p})[X]/(\bar{\mu})$, which are those that are generated by $\bar{\mu}_1, \dots, \bar{\mu}_g$. Their inverse images under ψ are $\mathfrak{p}_1, \dots, \mathfrak{p}_g$, and hence $\mathfrak{p}_1, \dots, \mathfrak{p}_g$ are exactly the prime ideals of B lying above $\mathfrak{p}$. Moreover, for each i , the field $k(\mathfrak{p}_i)$ is isomorphic to $k(\mathfrak{p})[X]/(\bar{\mu}_i)$, hence the residual degree of $\mathfrak{p}_i$ is $\deg \bar{\mu}_i$. Finally, we have the inclusion $\mathfrak{p}_1^{e_1} \dots \mathfrak{p}_g^{e_g} \subset \mathfrak{p}B$, hence $\mathfrak{p}B = \mathfrak{p}_1^{e'_1} \dots \mathfrak{p}_g^{e'_g}$ for some $e'_1 \leq e_1, \dots, e'_g \leq e_g$. But:

$$[L : K] = \sum_{i=1}^g e'_i f_i \leq \sum_{i=1}^g e_i f_i = \deg \bar{\mu} = [L : K].$$

Hence $e'_1 = e_1, \dots, e'_g = e_g$. □

Example 1.2.19. Let us come back to Example 1.1.22(i), so that $K = \mathbb{Q}$ and $L = \mathbb{Q}(x)$ with x such that $x^3 + x + 1 = 0$. We know that $\mathcal{O}_L = \mathbb{Z}[x]$. We then have:

1. $2\mathcal{O}_L = \mathfrak{p}_2$ prime with $e(\mathfrak{p}_2/(2)) = 1$ and $f(\mathfrak{p}_2/(2)) = 3$;
2. $3\mathcal{O}_L = \mathfrak{p}_3\mathfrak{p}_9$ with $\mathfrak{p}_3$ and $\mathfrak{p}_9$ distinct primes, $e(\mathfrak{p}_3/(3)) = e(\mathfrak{p}_9/(3)) = 1$, $f(\mathfrak{p}_3/(3)) = 1$ and $f(\mathfrak{p}_9/(3)) = 2$;
3. $31\mathcal{O}_L = \mathfrak{p}_{31}^2\mathfrak{p}'_{31}$ with $\mathfrak{p}_{31}$ and $\mathfrak{p}'_{31}$ distinct primes, $e(\mathfrak{p}_{31}/(31)) = 2$, $e(\mathfrak{p}'_{31}/(31)) = 1$ and $f(\mathfrak{p}_{31}/(31)) = f(\mathfrak{p}'_{31}/(31)) = 1$;
4. $59\mathcal{O}_L = \mathfrak{p}_{59}\mathfrak{p}'_{59}\mathfrak{p}''_{59}$ with $\mathfrak{p}_{59}$, $\mathfrak{p}'_{59}$ and $\mathfrak{p}''_{59}$ distinct primes, $e(\mathfrak{p}_{59}/(31)) = e(\mathfrak{p}'_{59}/(31)) = e(\mathfrak{p}''_{59}/(31)) = 1$ and $f(\mathfrak{p}_{59}/(31)) = f(\mathfrak{p}'_{59}/(31)) = f(\mathfrak{p}''_{59}/(31)) = 1$.

1.2.6 Ramification

As in the previous section, we consider the Framework 1.1.13. We are now ready to introduce some vocabulary to describe different ways prime factorizations behave under finite extensions:

Definition 1.2.20

Let $\mathfrak{p}$ be a non-zero prime ideal of A . Write the prime factorization of $\mathfrak{p}B$:

$$\mathfrak{p}B = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_g^{e_g}.$$

1. The prime ideal $\mathfrak{p}$ is said to be **nonsplit** in L if $g = 1$. It is said to be **split** otherwise, and **totally split** if $g = [L : K]$.
2. One of the prime ideals $\mathfrak{p}_i$ is said to be **unramified** in the extension L/K if $e_i = 1$ and the residue field extension $k(\mathfrak{p}_i)/k(\mathfrak{p})$ is separable. It is said to be **ramified** otherwise, and **totally ramified** if moreover $f_i = 1$.
3. The prime ideal $\mathfrak{p}$ is said to be **unramified** in L if all the $\mathfrak{p}_i$ are unramified. It is said **ramified** otherwise.

Example 1.2.21. In Example 1.2.19, the prime 2 is nonsplit and unramified, the prime 3 is split and unramified, the prime 31 is split and ramified, and the prime 59 is totally split (and unramified).

It turns out that prime ideals ramify only in exceptional situations:

Proposition 1.2.22

Almost all non-zero prime ideals of A are unramified in L .

Proof. Set $n := [L : K]$. Let α be a primitive element of L/K contained in B and let μ be its minimal polynomial. Set $B' = A[\alpha]$ and let $\mathfrak{p}$ be a non-zero prime ideal of A that is coprime to the discriminant $d := d(1, \alpha, \dots, \alpha^{n-1})$ and to the conductor $\mathfrak{f}(B/B')$. Since $d \notin \mathfrak{p}$, the reduction $\bar{\mu}$ of μ modulo $\mathfrak{p}$ has no multiple roots. We can therefore write $\bar{\mu} = \bar{\mu}_1 \dots \bar{\mu}_g$ with $\bar{\mu}_1, \dots, \bar{\mu}_g$ monic irreducible and pairwise distinct. By Proposition 1.2.18, the ideal $\mathfrak{p}B$ factors as $\mathfrak{p}_1 \dots \mathfrak{p}_g$ for some pairwise distinct prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_g$ with residual degrees $f_1 = \deg \bar{\mu}_1, \dots, f_g = \deg \bar{\mu}_g$. We deduce that the ramification indices of $\mathfrak{p}_1, \dots, \mathfrak{p}_g$ over K are all equal to 1 and that, for each i , the residue field $k(\mathfrak{p}_i)/k$ is generated by the class of α modulo $\mathfrak{p}_i$ and hence is separable. In other words, the prime ideal $\mathfrak{p}$ is unramified in L . $\square$

The previous proposition can be made much more precise thanks to the following theorem, that we will not prove in this course:

Theorem 1.2.23

Consider the **discriminant ideal** $\mathfrak{d}_{A/B}$, that is the ideal of A generated by the discriminants $d(\mathbf{x})$ of all K -bases $\mathbf{x}$ of L contained in B . A non-zero prime ideal in A ramifies in L if, and only if, it divides $\mathfrak{d}_{A/B}$.

Example 1.2.24. In Example 1.2.19, we have $\Delta_L = 31$, so 31 is the only prime number that ramified in L .

A particularly interesting situation arises when the extension L/K is Galois. We henceforth make that assumption and denote G the Galois group. We also keep the notations of Definition 1.2.20.

Proposition 1.2.25

The group G acts transitively on the set $\{\mathfrak{p}_1, \dots, \mathfrak{p}_g\}$ and we have $e_1 = \dots = e_g$ and $f_1 = \dots = f_g$.

Proof. First observe that:

$$\mathfrak{p}_1^{e_1} \dots \mathfrak{p}_g^{e_g} = \mathfrak{p}B = \sigma(\mathfrak{p}B) = \sigma(\mathfrak{p}_1)^{e_1} \dots \sigma(\mathfrak{p}_g)^{e_g} \quad (1.4)$$

for every $\sigma \in G$. This shows that G does act on $\{\mathfrak{p}_1, \dots, \mathfrak{p}_g\}$.

By contradiction, fix $2 \leq i_0 \leq g$ such that $\mathfrak{p}_{i_0}$ is not in the orbit of $\mathfrak{p}_1$. By the Chinese remainder Theorem, we can find $x \in \mathfrak{p}_{i_0}$ such that $x \equiv 1 \pmod{\sigma(\mathfrak{p}_1)}$ for all $\sigma \in G$. Then $N_{L/K}(x) = \prod_{\sigma \in G} \sigma(x)$ belongs to $\mathfrak{p}_{i_0} \cap A = \mathfrak{p}$ but not to $\mathfrak{p}_1$: contradiction! We deduce that G acts transitively on $\{\mathfrak{p}_1, \dots, \mathfrak{p}_g\}$.

Using equation (1.4) and the transitivity of the action, we deduce that $e_1 = \dots = e_g$, and that $k(\mathfrak{p}_1) \cong \dots \cong k(\mathfrak{p}_g)$, so that $f_1 = \dots = f_g$. $\square$

We will henceforth set $e = e_L(\mathfrak{p}) := e_1 = \dots = e_g$ and $f = f_L(\mathfrak{p}) := f_1 = \dots = f_g$.

Definition 1.2.26

For each $1 \leq i \leq g$, the stabilizer $\text{Stab}_{\text{Gal}(L/K)}(\mathfrak{p}_i)$ is called the **decomposition subgroup** of $\mathfrak{p}_i$ and is denoted $G_{\mathfrak{p}_i}$. The fixed field $M_i := L^{G_{\mathfrak{p}_i}}$ is called the **decomposition field** of $\mathfrak{p}_i$.

Proposition 1.2.27

Let C_i be the integral closure of A in M_i and set $q_i = p_i \cap B_i$.

1. The prime q_i is nonsplit in L and we have $e(p_i/q_i) = e$ and $f(p_i/q_i) = f$.
2. We have $e(q_i/p) = f(q_i/p) = 1$. In particular, the prime q_i is unramified in the extension M_i/K .

Proof. 1. We have $\text{Gal}(L/M_i) = G_{p_i}$. Hence $\sigma(p_i) = p_i$ for each $\sigma \in \text{Gal}(L/M_i)$. We deduce from Proposition 1.2.25 that p_i is the only ideal of B above q_i . Now we have $e(p_i/q_i) \leq e$ and $f(p_i/q_i) \leq f$. Using Propositions 1.2.17 and 1.2.25:

$$e(p_i/q_i)f(p_i/q_i) = [L : M_i] = |G_{p_i}| = \frac{[L : K]}{g} = ef \geq e(p_i/q_i)f(p_i/q_i).$$

We deduce that $e = e(p_i/q_i)$ and $f = f(p_i/q_i)$.

2. Part 1 together with the equalities $e(p_i/q_i)e(q_i/p) = e$ and $f(p_i/q_i)f(q_i/p) = f$ shows that $e(q_i/p) = f(q_i/p) = 1$.

□

Assume now, to simplify, that $k(p_i)$ is separable over $k(p)$. An element $\sigma \in G_{p_i}$ satisfies $\sigma(B) = B$ and $\sigma(p_i) = p_i$, and hence it naturally induces an element $\bar{\sigma} \in \text{Aut}(k(p_i)/k(p))$. We therefore get a group homomorphism:

$$\begin{aligned} \rho_i : G_{p_i} &\rightarrow \text{Aut}(k(p_i)/k(p)) \\ \sigma &\mapsto \bar{\sigma}. \end{aligned}$$

Proposition 1.2.28

The extension $k(p_i)/k(p)$ is Galois and the morphism ρ_i is surjective.

Proof. With the notations of Proposition 1.2.27, we know that $k(p) = k(q_i)$. We may therefore assume that $K = M_i$ and $D_{p_i} = \text{Gal}(L/K)$.

Now take some $\bar{\alpha} \in k(p_i)$ and consider a lifting $\alpha \in B$. Let μ be the minimal polynomial of α over K . Since L/K is Galois, we can write $\mu = (X - \alpha_1) \dots (X - \alpha_r)$ with $\alpha_1 = \alpha, \alpha_2, \dots, \alpha_r \in B$. Hence the reduction $\bar{\mu}$ of μ in $k(p)[X]$ splits as $(X - \bar{\alpha}_1) \dots (X - \bar{\alpha}_r)$ with $\bar{\alpha}_1, \dots, \bar{\alpha}_r \in k(p_i)$. This shows that all the conjugates of $\bar{\alpha} = \bar{\alpha}_1$ lie in $k(p_i)$, and hence $k(p_i)/k(p)$ is Galois.

Assume now that $\bar{\alpha}$ is a primitive element of $k(\mathfrak{p}_i)/k(\mathfrak{p})$ and take some $\bar{\sigma} \in \text{Gal}(k(\mathfrak{p}_i)/k(\mathfrak{p}))$. Then $\bar{\sigma}(\bar{\alpha}) = \bar{\alpha}_{i_0}$ for some i_0 . But we also know that there exists $\tau \in \text{Gal}(L/K)$ such that $\tau(\alpha) = \alpha_{i_0}$. Then $\rho_i(\tau)(\alpha) = \bar{\alpha}_{i_0} = \bar{\sigma}(\bar{\alpha})$, and hence $\rho_i(\tau) = \bar{\sigma}$. This settles the surjectivity of ρ_i . $\square$

Definition 1.2.29

The kernel of ρ_i is called the **inertia subgroup of $\mathfrak{p}_i$** and is denoted $I_{\mathfrak{p}_i}$. The corresponding fixed field $N_i := L^{I_{\mathfrak{p}_i}}$ is called the **inertia field of $\mathfrak{p}_i$** .

Proposition 1.2.30

We keep all the previous notations and assumptions. In particular, $k(\mathfrak{p}_i)$ is assumed to be separable over $k(\mathfrak{p})$. We let D_i be the integral closure of A in N_i and we set $\mathfrak{r}_i := \mathfrak{p}_i \cap D_i$.

1. We have $e(\mathfrak{p}_i/\mathfrak{r}_i) = e$ and $f(\mathfrak{p}_i/\mathfrak{r}_i) = 1$. In particular, the prime ideal $\mathfrak{r}_i$ is nonsplit and totally ramified in L/N_i .
2. We have $e(\mathfrak{r}_i/\mathfrak{q}_i) = 1$ and $f(\mathfrak{r}_i/\mathfrak{q}_i) = f$. In particular, the prime ideal $\mathfrak{q}_i$ is nonsplit and unramified in N_i/M_i .

Proof. By Proposition 1.2.27, we have $e(\mathfrak{p}_i/\mathfrak{r}_i)e(\mathfrak{r}_i/\mathfrak{q}_i) = e$ and $f(\mathfrak{p}_i/\mathfrak{r}_i)f(\mathfrak{r}_i/\mathfrak{q}_i) = f$. Moreover, by applying Proposition 1.2.28 to the extension L/N_i , we see that $\text{Gal}(k(\mathfrak{p}_i)/k(\mathfrak{r}_i))$ is trivial, so that $k(\mathfrak{p}_i) = k(\mathfrak{r}_i)$ and $f(\mathfrak{p}_i/\mathfrak{r}_i) = 1$. We deduce the equalities $e(\mathfrak{p}_i/\mathfrak{r}_i) = e$, $f(\mathfrak{r}_i/\mathfrak{q}_i) = f$ and $e(\mathfrak{r}_i/\mathfrak{q}_i) = 1$. $\square$

To summarize, we have decomposed our extension L/K as follows:

$$\begin{array}{ccccccc}
 \mathfrak{p}_i & \subset & B & \subset & L & & \\
 & & & & \downarrow & e(\mathfrak{p}_i/\mathfrak{r}_i)=e, f(\mathfrak{p}_i/\mathfrak{r}_i)=1, g_L(\mathfrak{r}_i)=1 & \\
 \mathfrak{r}_i & \subset & D_i & \subset & N_i & & \\
 & & & & \downarrow & e(\mathfrak{r}_i/\mathfrak{q}_i)=1, f(\mathfrak{r}_i/\mathfrak{q}_i)=f, g_{N_i}(\mathfrak{q}_i)=1 & \\
 \mathfrak{q}_i & \subset & C_i & \subset & M_i & & \\
 & & & & \downarrow & e(\mathfrak{q}_i/\mathfrak{p})=1, f(\mathfrak{q}_i/\mathfrak{p})=1, g_{M_i}(\mathfrak{p})=g & \\
 \mathfrak{p} & \subset & A & \subset & K & &
 \end{array}$$

1.3 The arithmetic behaviour of rings of integers

1.3.1 Minkowski's Theorem on lattices

In this section, we fix an n -dimensional Euclidean vector space V . We endow it with the corresponding Lebesgue measure, so that the volume of the cube:

$$\{t_1 e_1 + \cdots + t_n e_n : 0 \leq t_1, \dots, t_n < 1\}$$

spanned by an orthonormal basis $(e_1, \dots, e_n)$ of V is always 1. More generally, if $(v_1, \dots, v_n)$ is a basis of V , the parallelepiped spanned by $(v_1, \dots, v_n)$ is defined as:

$$\Pi_{v_1, \dots, v_n} := \{t_1 v_1 + \cdots + t_r v_r : 0 \leq t_1, \dots, t_r < 1\},$$

and its volume is the absolute value of the determinant of the transition matrix $P = (\langle e_i, v_j \rangle)_{1 \leq i, j \leq n}$ from some orthonormal basis $(e_1, \dots, e_n)$ of V to $(v_1, \dots, v_n)$. By observing that:

$$P^t P = \left(\sum_{k=1}^n \langle e_i, v_k \rangle \langle e_k, v_j \rangle \right)_{1 \leq i, j \leq n} = (\langle v_i, v_j \rangle)_{1 \leq i, j \leq n},$$

we get the orthonormal-basis-free formula:

$$\text{Vol}(\Pi_{v_1, \dots, v_n}) = \left(\det (\langle v_i, v_j \rangle)_{1 \leq i, j \leq n} \right)^{1/2}.$$

Definition 1.3.1

A **lattice** Γ in the n -dimensional Euclidean space V is a subgroup of the form:

$$\Gamma = \mathbb{Z}\gamma_1 + \cdots + \mathbb{Z}\gamma_r$$

where $\gamma_1, \dots, \gamma_r$ are linearly independent vectors in V . We then say that $(\gamma_1, \dots, \gamma_r)$ is a **basis** of Γ and that the set:

$$M := \{t_1 \gamma_1 + \cdots + t_r \gamma_r : 0 \leq t_1, \dots, t_r < 1\}$$

is a **fundamental parallelepiped** of Γ .

The lattice Γ is said to be **complete** if $r = n$. In that case, the volume of a fundamental parallelepiped M of Γ is independent of the choice of a basis of Γ . It is called the **covolume** of Γ and is denoted $\text{Covol}(\Gamma)$.

Example 1.3.2. If K is a quadratic imaginary extension of $\mathbb{Q}$, its ring of integers $\mathcal{O}_K$ is a complete lattice in $\mathbb{C}$.

Quite often, the most practical way to check that a subgroup of $\mathbb{R}^n$ is a lattice is provided by the following topological characterization:

Proposition 1.3.3

A subgroup of $\mathbb{R}^n$ is a lattice if, and only if, it is discrete.

Proof. The direct implication is obvious. Let us prove the converse. Let Γ be a non-zero discrete subgroup in $\mathbb{R}^n$. We can then construct inductively a finite sequence $\gamma_1, \gamma_2, \gamma_3, \dots$, of non-zero elements of Γ in the following way:

- for $i = 1$, we choose γ_1 to be a non-zero element of Γ with minimal norm;
- for $i > 1$, we set $V_i := \mathbb{R}\gamma_1 + \dots + \mathbb{R}\gamma_{i-1}$, we choose some $\gamma'_i \in \Gamma \setminus V_i$ if Γ is not contained in V_i and we let γ_i be an element of $\overline{\Pi}_{\gamma_1, \dots, \gamma_{i-1}, \gamma'_i} \cap \Gamma$ such that the distance $d(\gamma_i, V_i)$ is non-zero and minimal. Such an element exists because $\overline{\Pi}_{\gamma_1, \dots, \gamma_{i-1}, \gamma'_i}$ contains a finite number of elements of Γ .

The vectors $\gamma_1, \gamma_2, \gamma_3, \dots$ are linearly independent and hence the previous construction necessarily stops at some point. In other words, for some $r \geq 1$, we have $\Gamma \subset V_r$. Let us prove that $\Gamma = \bigoplus_{i=1}^r \mathbb{Z}\gamma_i$.

To do so, take some $\gamma \in \Gamma \setminus (\bigoplus_{i=1}^r \mathbb{Z}\gamma_i)$ and write $\gamma = x_1\gamma_1 + \dots + x_r\gamma_r$ with $x_1, \dots, x_r \in \mathbb{R}$. Let $s \in \{1, \dots, r\}$ be the smallest index such that $x_s \notin \mathbb{Z}$ and set:

$$\gamma' := \lfloor x_1 \rfloor \gamma_1 + \dots + \lfloor x_s \rfloor \gamma_s.$$

Then $x_1\gamma_1 + \dots + x_s\gamma_s - \gamma' = (x_s - \lfloor x_s \rfloor)\gamma_s \in \overline{\Pi}_{\gamma_1, \dots, \gamma_{s-1}, \gamma'_s} \cap \Gamma$ and:

$$d(x_1\gamma_1 + \dots + x_s\gamma_s - \gamma', V_s) = (x_s - \lfloor x_s \rfloor) \cdot d(\gamma_s, V_s) < d(\gamma_s, V_s).$$

This contradicts the definition of γ_s . We deduce that $\Gamma = \bigoplus_{i=1}^r \mathbb{Z}\gamma_i$, and hence Γ is a lattice. $\square$

The following corollary turns out to be extremely useful to prove that some subgroup of $\mathbb{R}^n$ is a complete lattice and to compute its covolume:

Corollary 1.3.4

Let Γ be a complete lattice in $\mathbb{R}^n$. Let $\Gamma' \subset \Gamma$ be a finite index subgroup.

Then Γ' is a complete lattice of $\mathbb{R}^n$ and:

$$\text{Covol}(\Gamma') = \text{Covol}(\Gamma) \cdot [\Gamma : \Gamma'].$$

Proof. Since Γ' is contained in Γ , it is discrete in $\mathbb{R}^n$, and hence it is a lattice by the previous proposition. The equality $\text{Covol}(\Gamma') = \text{Covol}(\Gamma) \cdot [\Gamma : \Gamma']$ follows immediately from the definition of the covolume and the Smith normal form Theorem. $\square$

We are now ready to move on to the so-called Minkowski's theory. The intuitive idea is that, if one takes a complete lattice Γ in $\mathbb{R}^n$ and some "sufficiently large" subset $X \subset \mathbb{R}^n$, then X has to contain some elements of Γ . We start with the following proposition:

Proposition 1.3.5 (Blichfeldt)

Let Γ be a complete lattice in the vector space $V = \mathbb{R}^n$, and let X be a measurable subset of V such that:

$$\text{Vol}(X) > \text{Covol}(\Gamma).$$

Then there exist two elements $x_1, x_2 \in X$ such that $x_2 - x_1 \in \Gamma \setminus \{0\}$.

Proof. Recall that M is a fundamental parallelepiped. We have:

$$V = \bigcup_{\gamma \in \Gamma} (\gamma + M),$$

hence:

$$X = \bigcup_{\gamma \in \Gamma} ((\gamma + M) \cap X).$$

We deduce that:

$$\text{Vol}(M) < \text{Vol}(X) \leq \sum_{\gamma \in \Gamma} \text{Vol}((\gamma + M) \cap X) = \sum_{\gamma \in \Gamma} \text{Vol}(M \cap (X - \gamma)).$$

Therefore there exist $\gamma_1, \gamma_2 \in \Gamma$ with $\gamma_1 \neq \gamma_2$ and:

$$M \cap (X - \gamma_1) \cap (X - \gamma_2) \neq \emptyset.$$

We can therefore find $x_1, x_2 \in X$ such that $x_1 - \gamma_1 = x_2 - \gamma_2$, and hence $x_2 - x_1 = \gamma_2 - \gamma_1 \in \Gamma$. $\square$



Hans Frederick Blichfeldt (Illar 1873 - Palo Alto 1945)

Theorem 1.3.6 (Minkowski)

Let Γ be a complete lattice in the vector space $V = \mathbb{R}^n$, and let X be a (measurable) centrally symmetric convex subset of V such that:

$$\text{Vol}(X) > 2^n \text{Covol}(\Gamma).$$

Then X contains a non-zero element of Γ .

Proof. Set $Y := \frac{1}{2}X$, so that $\text{Vol}(Y) > \text{Covol}(\Gamma)$. By Proposition 1.3.5, there exist $y_1, y_2 \in Y$ such that $y_2 - y_1 \in \Gamma \setminus \{0\}$. Moreover, since X is centrally symmetric and convex:

$$y_2 - y_1 = \frac{(-2y_1) + (2y_2)}{2} \in X.$$

We have therefore found a non-zero element of Γ in X . □



Hermann Minkowski (Aleksotas 1864 - Göttingen 1909)

1.3.2 Geometry of numbers

Let K be a number field and consider the set of complex field embeddings $\Sigma := \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$. This set has $n := [K : \mathbb{Q}]$ elements and it is naturally endowed with an action of $\text{Gal}(\mathbb{C}/\mathbb{R})$ by complex conjugation. The subset of fixed points for this action is the set of real embeddings $\Sigma_{\mathbb{R}} := \text{Hom}_{\mathbb{Q}}(K, \mathbb{R})$, whose cardinal will be denoted r . Its complement $\Sigma \setminus \Sigma_{\mathbb{R}}$ is partitioned into a certain number s of equivalence classes, each of them of cardinal 2. We let $\Sigma_{\mathbb{C}}$ be a set of representatives of those equivalence classes in $\Sigma \setminus \Sigma_{\mathbb{R}}$, so that:

$$\Sigma = \Sigma_{\mathbb{R}} \sqcup \Sigma_{\mathbb{C}} \sqcup \overline{\Sigma_{\mathbb{C}}}, \quad n = r + 2s.$$

We introduce the embedding:

$$\begin{aligned} \iota : K &\rightarrow \mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}} \\ x &\mapsto (\sigma(x))_{\sigma \in \Sigma_{\mathbb{R}} \sqcup \Sigma_{\mathbb{C}}}. \end{aligned}$$

Proposition 1.3.7

The image of $\mathcal{O}_K$ by ι through the embedding ι is a complete lattice in the n -dimensional real vector space $\mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}}$.

Proof. According to Proposition 1.3.3, in order to check that $\iota(\mathcal{O}_K)$ is a lattice, we only need to prove that it is discrete. To do so, consider a sequence $(x_n)_{n \geq 0}$ of elements in $\mathcal{O}_K$ such that the sequence $(\iota(x_n))_{n \geq 0}$ converges to $\iota(x)$ for some

$x \in \mathcal{O}_K$. We prove that $(x_n)_{n \geq 0}$ is eventually constant equal to x . Indeed, for any $\sigma \in \text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$, we have $\sigma(x_n) \xrightarrow{n \rightarrow \infty} \sigma(x)$. Hence:

$$\chi_{K/\mathbb{Q}, x_n} \xrightarrow{n \rightarrow \infty} \chi_{K/\mathbb{Q}, x}$$

coefficient-wise. But all the x_n 's and x belong to $\mathcal{O}_K$, so that all the $\chi_{K/\mathbb{Q}, x_n}$'s and $\chi_{K/\mathbb{Q}, x}$ have integer coefficients. We deduce that $\chi_{K/\mathbb{Q}, x_n} = \chi_{K/\mathbb{Q}, x}$, and hence that x_n is a root of $\chi_{K/\mathbb{Q}, x}$, for n large enough. In particular, $(x_n)_{n \geq 0}$ is eventually constant equal to x , as wished. Since $\mathcal{O}_K$ has rank n and $\mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}}$ has dimension n as a real vector space, we conclude that $\iota(\mathcal{O}_K)$ is a complete lattice in $\mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}}$. $\square$

Proposition 1.3.8

Endow the real vector space $\mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}}$ with the scalar product:

$$\langle (x_{\sigma})_{\sigma \in \Sigma_{\mathbb{R}}}, (z_{\sigma})_{\sigma \in \Sigma_{\mathbb{C}}} \rangle, \langle (x'_{\sigma})_{\sigma \in \Sigma_{\mathbb{R}}}, (z'_{\sigma})_{\sigma \in \Sigma_{\mathbb{C}}} \rangle = \sum_{\sigma \in \Sigma_{\mathbb{R}}} x_{\sigma} x'_{\sigma} + 2\text{Re} \left(\sum_{\sigma \in \Sigma_{\mathbb{C}}} z_{\sigma} \overline{z'_{\sigma}} \right).$$

The covolume of $\iota(\mathcal{O}_K)$ is then equal to $\sqrt{|\Delta_K|}$.

Proof. Consider a basis $(e_1, \dots, e_n)$ of K such that $\mathcal{O}_K = \mathbb{Z}e_1 + \dots + \mathbb{Z}e_n$. We then have:

$$\begin{aligned} \text{Covol}(\iota(\mathcal{O}_K))^2 &= \det(\langle \iota(e_i), \iota(e_j) \rangle)_{1 \leq i, j \leq n} \\ &= \det \left(\sum_{\sigma \in \Sigma_{\mathbb{R}}} \sigma(e_i e_j) + \sum_{\sigma \in \Sigma_{\mathbb{C}}} \sigma(e_i) \overline{\sigma(e_j)} + \sum_{\sigma \in \Sigma_{\mathbb{C}}} \overline{\sigma(e_i)} \sigma(e_j) \right)_{1 \leq i, j \leq n} \\ &= \det \left(\sum_{\sigma \in \Sigma} \sigma(e_i) \overline{\sigma(e_j)} \right)_{1 \leq i, j \leq n}. \end{aligned}$$

Observing that :

$$\left(\sum_{\sigma \in \Sigma} \sigma(e_i) \overline{\sigma(e_j)} \right)_{1 \leq i, j \leq n} = A^t \overline{A}$$

with $A = (\sigma(e_i))_{\sigma \in \Sigma, 1 \leq i \leq n}$, we get:

$$\text{Covol}(\iota(\mathcal{O}_K))^2 = |\det A|^2 = |\Delta_K|.$$

$\square$

Corollary 1.3.9

Keeping the previous conventions and notations, if I is a non-zero ideal of $\mathcal{O}_K$, then $\iota(I)$ is a complete lattice in the n -dimensional real vector space $\mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}}$ with covolume $[\mathcal{O}_K : I] \cdot \sqrt{|\Delta_K|}$.

Proof. By Proposition ??, the ideal I has finite index in $\mathcal{O}_K$. The result then follows by combining Propositions 1.3.7 and 1.3.8 with Corollary 1.3.4.4. $\square$

Theorem 1.3.10

Let I be a non-zero ideal in $\mathcal{O}_K$. Fix $(c_\sigma)_{\sigma \in \Sigma} \in \mathbb{R}_+^{\Sigma}$ such that $c_{\bar{\sigma}} = c_\sigma$ for all $\sigma \in \Sigma$ and:

$$\prod_{\sigma \in \Sigma} c_\sigma > \left(\frac{2}{\pi}\right)^s \cdot [\mathcal{O}_K : I] \cdot \sqrt{|\Delta_K|}.$$

There exists a non-zero element $x \in I$ such that $|\sigma(x)| < c_\sigma$ for all $\sigma \in \Sigma$.

Proof. Consider the set:

$$X := \prod_{\sigma \in \Sigma_{\mathbb{R}}} (-c_\sigma, c_\sigma) \times \prod_{\sigma \in \Sigma_{\mathbb{C}}} D(0, c_\sigma) \subset \mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}},$$

where $D(0, c_\sigma)$ stands for the disk of centre 0 and radius c_σ in $\mathbb{C}$. It is of course a centrally symmetric convex subset of $\mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{C}^{\Sigma_{\mathbb{C}}}$, and:

$$\begin{aligned} \text{Vol}(X) &= \prod_{\sigma \in \Sigma_{\mathbb{R}}} (2c_\sigma) \cdot \prod_{\sigma \in \Sigma_{\mathbb{C}}} (2\pi c_\sigma^2) \\ &= 2^{r+s} \cdot \pi^s \cdot \prod_{\sigma \in \Sigma} c_\sigma \\ &> 2^{r+s} \cdot \pi^s \cdot \left(\frac{2}{\pi}\right)^s \cdot [\mathcal{O}_K : I] \cdot \sqrt{|\Delta_K|} \\ &> 2^n \cdot \text{Covol}(\iota(I)). \end{aligned}$$

By Theorem 1.3.6, we deduce that X contains some non-zero element of the lattice $\iota(I)$. In other words, there exists a non-zero element $x \in I$ such that $|\sigma(x)| < c_\sigma$ for all $\sigma \in \Sigma$. $\square$

Corollary 1.3.11

For any non-zero ideal I in $\mathcal{O}_K$, there exists a non-zero element $x \in I$ such

that:

$$|N_{K/\mathbb{Q}}(x)| \leq \left(\frac{2}{\pi}\right)^s \cdot [\mathcal{O}_K : I] \cdot \sqrt{|\Delta_K|}.$$

Proof. Given $c > \left(\frac{2}{\pi}\right)^s \cdot [\mathcal{O}_K : I] \cdot \sqrt{|\Delta_K|}$, choose a family $(c_\sigma)_{\sigma \in \Sigma} \in \mathbb{R}_+^\Sigma$ such that $c_{\bar{\sigma}} = c_\sigma$ for all $\sigma \in \Sigma$ and:

$$\prod_{\sigma \in \Sigma} c_\sigma = c.$$

According to Theorem 1.3.10, there exists a non-zero element $x_c \in I$ such that $|\sigma(x_c)| < c_\sigma$ for all $\sigma \in \Sigma$. In particular:

$$|N_{K/\mathbb{Q}}(x_c)| = \prod_{\sigma \in \Sigma} |\sigma(x_c)| < \prod_{\sigma \in \Sigma} c_\sigma = c.$$

The corollary immediately follows since $N_{K/\mathbb{Q}}(x_c)$ always belongs to $\mathbb{Z}$ and since the previous inequality holds for any $c > \left(\frac{2}{\pi}\right)^s \cdot [\mathcal{O}_K : I] \cdot \sqrt{|\Delta_K|}$. $\square$

1.3.3 Finiteness of the ideal class group

Definition 1.3.12

Let A be a Dedekind domain with fraction field K . We denote by $\mathcal{I}_K$ the set of non-zero fractional ideals of A and by $\mathcal{P}_K \subset \mathcal{I}_K$ the subset consisting of those fractional ideals that are **principal**, that is of the form xA for some $x \in K^\times$. According to Corollary 1.2.9.2, $\mathcal{I}_K$ is an abelian group, and $\mathcal{P}_K$ is a subgroup. The **ideal class group** of K is defined as $\text{Cl}_K := \mathcal{I}_K / \mathcal{P}_K$.

In this section, we aim at proving that the ideal class group Cl_K of a number field is always finite. To do so, we will apply Corollary 1.3.11.5 to find in each class $c \in \text{Cl}_K$ an ideal $I \in c$ contained in $\mathcal{O}_K$ and with bounded index $[\mathcal{O}_K : I]$. It will therefore be useful to understand how this index behaves:

Lemma 1.3.13. *Let K be a number field.*

- (i) *For any non-zero element $a \in \mathcal{O}_K$, we have $[\mathcal{O}_K : (a)] = |N_{K/\mathbb{Q}}(a)|$.*
- (ii) *For any ideals I and J in $\mathcal{O}_K$, we have $[\mathcal{O}_K : IJ] = [\mathcal{O}_K : I] \cdot [\mathcal{O}_K : J]$.*

Proof. (i) This is a direct consequence of Lemma ??.

- (ii) We have to prove that, if $I = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r}$ is the prime decomposition of some ideal I of $\mathcal{O}_K$, then:

$$[\mathcal{O}_K : I] = \prod_{i=1}^r [\mathcal{O}_K : \mathfrak{p}_i]^{e_i}.$$

For that purpose, we first observe that, according to the Chinese remainder Theorem:

$$\mathcal{O}_K/I \cong \prod_{i=1}^r \mathcal{O}_K/\mathfrak{p}_i^{e_i},$$

and hence:

$$[\mathcal{O}_K : I] = \prod_{i=1}^r [\mathcal{O}_K : \mathfrak{p}_i^{e_i}].$$

Now fix some $i \in \{1, \dots, r\}$ and some $s \geq 0$. Since $\mathfrak{p}_i^s \neq \mathfrak{p}_i^{s+1}$, the quotient $\mathfrak{p}_i^s/\mathfrak{p}_i^{s+1}$ is a non-trivial $\mathcal{O}_K/\mathfrak{p}_i$ -vector space. Choose some $x \in \mathfrak{p}_i^s \setminus \mathfrak{p}_i^{s+1}$, and consider the $\mathcal{O}_K/\mathfrak{p}_i$ -linear map:

$$\begin{aligned} \varphi_s : \mathcal{O}_K/\mathfrak{p}_i &\rightarrow \mathfrak{p}_i^s/\mathfrak{p}_i^{s+1} \\ y &\mapsto xy. \end{aligned}$$

Since $x \in \mathfrak{p}_i^s \setminus \mathfrak{p}_i^{s+1}$, we have $(x) + \mathfrak{p}_i^{s+1} = \mathfrak{p}_i^s$, and hence φ_s is surjective. We deduce that φ_s is an isomorphism, so that:

$$[\mathcal{O}_K : \mathfrak{p}_i] = [\mathfrak{p}_i^s : \mathfrak{p}_i^{s+1}].$$

This being true for all s , we deduce that:

$$[\mathcal{O}_K : \mathfrak{p}_i^{e_i}] = [\mathcal{O}_K : \mathfrak{p}_i]^{e_i},$$

and hence:

$$[\mathcal{O}_K : I] = \prod_{i=1}^r [\mathcal{O}_K : \mathfrak{p}_i]^{e_i}.$$

□

Definition 1.3.14

The index $[\mathcal{O}_K : I]$ is called the **norm** of I and is denoted $N(I)$.

Proposition 1.3.15

For any ideal class $c \in \text{Cl}_K$, there exists a fractional ideal $I \in c$ that is contained in $\mathcal{O}_K$ and such that:

$$N(I) \leq \left(\frac{2}{\pi}\right)^s \cdot \sqrt{|\Delta_K|}.$$

Proof. Choose a fractional ideal $I_0 \in \mathcal{I}_K$ representing c and choose an element $a \in \mathcal{O}_K \setminus \{0\}$ such that $J_0 := aI_0^{-1}$ is an ideal in $\mathcal{O}_K$. According to Corollary 1.3.11.5, there exists a non-zero element $x \in J_0$ such that:

$$|N_{K/\mathbb{Q}}(x)| \leq \left(\frac{2}{\pi}\right)^s \cdot [\mathcal{O}_K : J_0] \cdot \sqrt{|\Delta_K|}.$$

Since $I_0 J_0 = (a)$, Lemma 1.3.13 implies that:

$$N(J_0) = \frac{|N_{K/\mathbb{Q}}(a)|}{N(I_0)},$$

and hence, noting that $xa^{-1}I_0$ is an ideal in $\mathcal{O}_K$:

$$N(xa^{-1}I_0) = \frac{|N_{K/\mathbb{Q}}(x)|}{|N_{K/\mathbb{Q}}(a)|} \cdot N(I_0) \leq \left(\frac{2}{\pi}\right)^s \cdot \sqrt{|\Delta_K|}.$$

We conclude by setting $I := xa^{-1}I_0$. □

Remark 1.3.16. It is in fact possible to improve the previous upper bound to $\frac{n!}{n^n} \left(\frac{4}{\pi}\right)^s \sqrt{|\Delta_K|}$.

Theorem 1.3.17

For any number field K , the ideal class group Cl_K is finite.

Proof. For any $M \geq 1$, there exist finitely many ideals I in $\mathcal{O}_K$ satisfying the inequality $N(I) \leq M$. The Theorem is therefore a direct application of Proposition 1.3.15. □

1.3.4 Dirichlet's Unit Theorem

Let K be a number field of degree n . Let $r = |\Sigma_{\mathbb{R}}|$ and $s = |\Sigma_{\mathbb{C}}|$, so that $n = r + 2s$. We aim at understanding the group $\mathcal{O}_K^\times$. To do so, we consider the morphism:

$$\begin{aligned} \lambda : K^\times &\rightarrow \mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{R}^{\Sigma_{\mathbb{C}}} \\ z &\mapsto ((\log |\sigma(z)|))_{\sigma \in \Sigma_{\mathbb{R}}}, (2 \log |\sigma(z)|)_{\sigma \in \Sigma_{\mathbb{C}}}. \end{aligned}$$

Lemma 1.3.18. 1. The image of $\mathcal{O}_K^\times$ is contained in:

$$V := \{(x_\sigma)_{\sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}} \in \mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{R}^{\Sigma_{\mathbb{C}}} : \sum_{\sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}} x_\sigma = 0\}.$$

2. Every bounded subset of $\lambda(\mathcal{O}_K^\times)$ is finite. In particular, $\ker(\lambda)$ is a finite group equal to the group $\mu(K)$ of roots of unity of K .

Proof. 1. Take $z \in \mathcal{O}_K^\times$. Then $N_{K/\mathbb{Q}}(z) \in \mathbb{Z}^\times = \{\pm 1\}$. Hence:

$$\sum_{\sigma \in \Sigma_{\mathbb{R}}} \log |\sigma(z)| + 2 \sum_{\sigma \in \Sigma_{\mathbb{C}}} \log |\sigma(z)| = \log \left(\prod_{\sigma \in \Sigma} |\sigma(z)| \right) = 0,$$

and $\lambda(z) \in V$.

2. Now take some $R > 0$. Given $z \in \Gamma \cap B(0, R)$, all the conjugates of z have modulus at most e^R , and hence there exists an integer $M > 0$ independent of z such that the coefficients of the minimal polynomial μ_z of z are bounded by M . We deduce that $\Gamma \cap B(0, R)$ is finite. In particular, $\ker(\lambda)$ is a finite group and has to be contained in $\mu(K)$. The reverse inclusion is obvious. □

We henceforth set $\Gamma := \lambda(\mathcal{O}_K^\times)$.

Proposition 1.3.19

The group Γ is a complete lattice in V .

Before proving this proposition, we need a small lemma:

Lemma 1.3.20. *For each $C > 0$, there exists a finite subset $\Theta(C) \subset \mathcal{O}_K \setminus \{0\}$ such that every non-zero element $z \in \mathcal{O}_K$ with $|N_{K/\mathbb{Q}}(z)| \leq C$ is associated to an element of $\Theta(C)$.*

Proof. Fix some positive integer $c < C$. We claim that, in each class of $\mathcal{O}_K$ modulo $c\mathcal{O}_K$, there is at most one element with norm $\pm c$. Indeed, if α and β are two elements with norm $\pm c$ lying in the same class, we can write $\alpha = \beta + c\gamma$ for some $\gamma \in \mathcal{O}_K$, and then:

$$\begin{aligned} \frac{\alpha}{\beta} &= 1 + \frac{c}{\beta}\gamma = 1 \pm \frac{N_{K/\mathbb{Q}}(\beta)}{\beta}\gamma \in \mathcal{O}_K, \\ \frac{\beta}{\alpha} &= 1 - \frac{c}{\alpha}\gamma = 1 \pm \frac{N_{K/\mathbb{Q}}(\alpha)}{\alpha}\gamma \in \mathcal{O}_K. \end{aligned}$$

The lemma follows from the finiteness of the quotient $\mathcal{O}_K / c\mathcal{O}_K$. □

Proof of Prop. 1.3.19. By Lemma 1.3.18, the group Γ is a discrete subgroup of V . By Proposition 1.3.3, it is a lattice in V .

Now consider a family $(c_\sigma)_{\sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}} \in \mathbb{R}_{>0}^{\Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}}$ such that:

$$C := \prod_{\sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}} c_\sigma > \left(\frac{2}{\pi}\right)^s \sqrt{|\Delta_K|},$$

and introduce the set:

$$X := \{(x_\sigma)_{\sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}} \in \mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{R}^{\Sigma_{\mathbb{C}}} : \forall \sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}, x_\sigma < \log c_\sigma\}.$$

For any $\nu \in V$, we have:

$$X - \nu = \{(x_\sigma)_{\sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}} \in \mathbb{R}^{\Sigma_{\mathbb{R}}} \times \mathbb{R}^{\Sigma_{\mathbb{C}}} : \forall \sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}, x_\sigma < \log c'_\sigma\}$$

for some $c'_\sigma > 0$ such that:

$$\prod_{\sigma \in \Sigma_{\mathbb{R}} \cup \Sigma_{\mathbb{C}}} c'_\sigma \cdot \prod_{\sigma \in \Sigma_{\mathbb{C}}} c'_\sigma = C.$$

By Theorem 1.3.10, there exists $a_\nu \in \mathcal{O}_K \setminus \{0\}$ such that $\lambda(a_\nu) \in X - \nu$. The element a_ν satisfies:

$$|N_{K/\mathbb{Q}}(a_\nu)| < C.$$

Hence, by Lemma 1.3.20, there exists some $b_\nu \in \Theta(C)$ such that a_ν and b_ν are associated. This being true for all $\nu \in V$, we deduce that:

$$V = \lambda(\mathcal{O}_K^\times) + Y$$

with:

$$Y = V \cap \bigcup_{b \in \Theta(C)} (X - \lambda(b)).$$

But $V \cap (X - \lambda(b))$ is bounded for each $b \in \Theta(C)$, hence so is Y . We deduce that $\lambda(\mathcal{O}_K^\times)$ is a complete lattice in V . $\square$

Theorem 1.3.21

Let K be a number field of degree n . Let $r = |\Sigma_{\mathbb{R}}|$ and $s = |\Sigma_{\mathbb{C}}|$, so that $n = r + 2s$. Let $\mu(K)$ be the group de roots of unity inside K . Then $\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r+s-1}$, with $\mu(K)$ finite.

Proof. By Lemma 1.3.18, we have an exact sequence:

$$1 \rightarrow \mu(K) \rightarrow \mathcal{O}_K^\times \rightarrow \lambda(\mathcal{O}_K^\times) \rightarrow 0,$$

in which $\mu(K)$ is finite. Moreover, by Proposition 1.3.19, the group $\lambda(\mathcal{O}_K^\times)$ is free abelian of rank $\dim V = r + s - 1$. Hence the previous exact sequence splits and $\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r+s-1}$. $\square$



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